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CHM 121 — Past Questions & Materials
by Renance

Course: CHM 121
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 100

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Paper 1 of 2 — CHM121 organic chemistry

Level 100 • Second Semester • PastQuestion

| r 4 ee OPTION A

ye FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,

- DEPARTMENT OF CHEMISTRY

SECOND SEMESTER EXAMINATIONS 2023 / 2024 SESSION

| | CHM 121 ORGANIC CHEMISTRY TIME: 1 Hr Answer all questions

\ Shade your Correct Option on the Answer Sheet

i j

| 1. Carbon nanotubes are composed of: 11. ----- is a method useful to determine conjugated

} (a) Layers of graphene rolled into tubes (b) Sheets system (a) NMR (b) IR (c) UV Spectroscopy (4) X-ray

: of diamond (c) Chains of carbon atoms (d)

i | Amorphous carbon clusters 12. The term organic was originally given because

organic compounds were believed to be produced only

p | 2. Which property makes carbon nanotubes suitable by ----- (a) living organisms (b) micro organisms (c) -

2 for nanotechnology applications? macro organisms (d) microbial particles

2) (a) High density (b) High tensile strength

mie c) Low electrical conductivity d) Low 13. One of the following is not a reason why carbon melting point forms such a large number of compounds compared to

} other elements (a) its forms ring structure (b) its having

3. Which of the following describes the bonding in the maximum covalency of four and it has a half-filled

carbon nanotubes? (a) sp³ hybridization L-Shell (c) it forms single, double and triple bonds with (b) sp² hybridization (c) Ionic bonding other carbon atoms or other atoms (d) crystallization

(d) Metallic bonding

14. Organic compounds are classified into the

4, In a covalent bond, electrons are: following major groups except one (a) aliphatic (b)

(a) Transferred (b) Shared (c) Lost aromatic (c) amorphous (d) heterocyclic

(d) Gained

15. One of the following is not a method of isolating

5. A pi (x) bond is formed by: (a) End-to-end overlap and purifying organic compounds (a) sublimation (b)

of orbitals (b) Side-to-side overlap of p orbitals atomization (c) distillation (d) chromatography

(c) Overlap of s orbitals (d) No orbital interaction

16. The method of separation that involves stationary'

6. Which of the following compounds exhibits sp^2 hybridization? (a) chromatography (b) zone melting (c) sublimation (d) distillation
7. Which molecule contains only sigma bonds? (a) CH_4 (b) C_2H_4 (c) C_2H_2 (d) C_2H_6
8. Who was one of the key scientists involved in the discovery of fullerenes? (a) empirical formula (b) normality (c) molecular formula (d) concentration
9. Inductive effect is due to difference in (a) bond angle (b) bond length (c) electronegativity (d) dipole moment
10. To detect nitrogen, the filtered aqueous alkaline of the mixture (a) remove interference, adjust the pH, solution obtained by sodium fusion is heated with (b) iron (II) sulphate (c) iron (III) sulphate (d) copper (II) sulphate
11. Which of the following distillation methods is not used to separate azeotropic mixtures? (a) azeotropic distillation (b) destructive distillation (c) pressure swing distillation (d) reactive distillation
12. Reaction $CaC_2 + H_2O \longrightarrow$ (a) ethene (b) methane (c) ethane (d) acetylene
13. The reagent that can differentiate 2-butyne and 2-butene is (a) $KMnO_4$ (b) Br_2 (c) $NaOH$ (d) $NaCl$
14. The compound which is not a ketone is (a) acetone (b) butan-2-one (c) pentan-3-one (d) hexan-4-one
15. The compound which is not a primary amine is (a) ethylamine (b) propylamine (c) butylamine (d) dimethylamine
16. The compound which is not a secondary amine is (a) diethylamine (b) dipropylamine (c) dibutylamine (d) triethylamine
17. One of the following is not a physical property (a) colour (b) odour (c) density (d) xylene
18. The simplest relative number of each kind of atom in the molecule is called (a) empirical formula (b) molecular formula (c) normality (d) concentration
19. In the process of determining sulphur content in a sodium fusion sample, it is recommended to acidify the solution obtained by sodium fusion is heated with (a) iron (II) sulphate (b) iron (III) sulphate (c) copper (II) sulphate (d) copper (I) sulphate
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23. The compound which is not a secondary amine is (a) diethylamine (b) dipropylamine (c) dibutylamine (d) triethylamine
24. The compound which is not a tertiary amine is (a) triethylamine (b) tripropylamine (c) tributylamine (d) trimethylamine
25. The compound which is not a quaternary ammonium salt is (a) $(CH_3)_4N^+Cl^-$ (b) $(CH_3)_3N^+CH_3Cl^-$ (c) $(CH_3)_2N^+CH_3CH_3Cl^-$ (d) $(CH_3)_2N^+CH_3CH_2CH_3Cl^-$
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22. Most compounds with chiral atoms do not have a plane of symmetry. However, a compound with chiral atoms and a plane of symmetry is called a ____ (a) mesoform (b) diastereomer (c) tautomer (d) cis or trans-isomer
23. The reagent typically used to detect halogens in the sodium fusion test is (a) silver nitrate (b) allyl chloride (c) ethenyl chloride (d) ethene chloride
24. What is the primary purpose of IR spectroscopy is to ____ (a) determine molecular weight (b) identify functional groups in a molecule (c) analyze elemental composition (d) measure electrical conductivity
25. Isomerism is the ____ (a) ability of molecules to conduct electricity (b) existence of compounds with the same molecular formula but different structures or arrangements (c) process of molecular bonding (d) classification of elements in the periodic table
26. Which type of isomerism involves different spatial arrangements of atoms in a molecule? (a) structural isomerism (b) geometric isomerism (c) optical isomerism (d) tautomeric isomerism
27. Which of the following is true of the percentage composition of element in the compound (a) Benzene (b) Propanone (c) Phenyl acetone (d) Acetophenone
28. 1g of a compound X containing C, H and O by reaction with KMnO_4 in H_2SO_4 gave 2.442g of CO_2 and 1.00g of H_2O . If the compound (a) is a ketone (b) acidifies water (c) is a primary alcohol (d) is a secondary alcohol
29. Propene can be made by (a) hydration of ethene (b) cracking of ethane (c) cracking of propene (d) hydration of propene
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32. Give one example of a compound with a chiral atom and a plane of symmetry (a) Br_2/CCl_4 (b) $\text{AgNO}_3/\text{NH}_4\text{OH}$ (c) CH_3Br (d) $\text{C}_2\text{H}_5\text{SO}_4$
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molecular weight of the compound is 72, calculate the (c) BH₃; NaOH (d) Mercuric acid molecular formula. (a) CsHeO₄

(b) CaHsO (c) CoH₂₀₆ (4) CoHcO 40 Give the name of the following compound < OH

29. Which compound is commonly referred to as MARSH gas? (a) Propane (b) propene (c) OX methane (d) butane ee el

2n/ 1 (a) m-chlorophenol —(b) ; —. —> m-hydroxy chlorobenzene

30 (CH₃)₂C = CHCH₂CH₃ + O₃ ? (c) Chloroform (d) 4 cHloronheaet y

A + B; A and B in the above reaction are

(a) (CH₃)₂CH₂COOH + CH₃;COOH

(b) (CH₃)₂CH₂CHO + CH₃CHO

(c) (CH₃)₂CHO + CH₃;CH₂.CHO |

OPTION A

48. The dehydration

of alcohols generally proceeds

The initiation step in free radical chlorination

of . . “eames ae : via which type of mechanism?

M V (a) Sn₂ (b) E₁ (c) E₂ (d) Sn¹

(a) Qh. —=-- 2c * (b) Cha > CH₃* + Ht

hy 49. Which of the following is a primary amine?

ic) CH₃Cl pie” CH₃* + Cl* (a). Methanamide (b). Dimethylamine (c)

(d) None of the above Trimethylamine (4) Aniline

426 1-chloropropane is converted to propene by 50. Which of the following reactions is used to reacting it with ----- (a) NaOH / alcohol prepare primary amines?

(a) Gabriel phthalimide synthesis (b) Hoffman kylation

(c)Na () KMnO₄,

degradation (c) Friedel-Crafts al

(b) ZnCl₂

(d) Wurtz reaction

43 Give the IUPAC name of the compound whose structure is

Br Br

nitrous acid?

azonium salt (b) It forms a nitro

SS COOH (a) It forms a di

compound (c) It forms a hydrocarbon

(d) It forms an alcohol

51. What happens when a primary amine reacts with

5-ethyl-2-propyl-3-methyl-4-nitrobenzene

classified into primary.

depending on the number of

(b) Hydroxyl groups

to the carbon bonded to —

: : 52. Alcohols can be

9-diethyl-3,3-dimethyl-5- secondary, and tertiary depending

9 (a) Oxygen atoms

(a) 4,8-dibromo-3,3-diethyl

undecenoic acid

(b) 4,8-dibromo-6,

dodecenoic acid

(c) 4,8-dibromo-6,9-diethyl-3,3-dimethyl-6- (©) Alkyl groups attached

dodecenoic acid : OH (d) Carbons in the chain

(d) 4,8-dibromo-3,3-diethyl-6-ethyl-9-propyl-6-

53. The process of converting sugars into alcohol is

ysis (b) Fermentation (c)

undecenoic acid

called? (a) Hydrolysis

Hydration (d) Saponification

44 $\text{CH}_3\text{CH}_2\text{CHCH}_2\text{CHOHCH}_3$

novamaltol?

54, Which of the following is

(a) Phenol (d) Ethanol

i

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$

(a) Methanol (b) Glycerol (c)

(a) 2-hexyl-3-hexanol (b) 3-hexyl-2-hexanol

hexanol (c) 4-ethyl-9-methyl-2-decanol

(d) 7-ethyl-2-methyl-9-decanol 55. Which alcohol is known to cause blindness if

ingested? (a) Ethanol (b) Methanol (c)

1-propanol (d) Butanol

of the following is the major product

45, Which

ethanol is heated with concentrated sulphuric acid at

170°C? (a). Ethene 56, What type of reaction is the conversion of alcohol to alkene? (a) Substitution

(d) Polymerization

Aldehyde (d) Acetic acid

(b) Addition (c) Elimination

(b) Ether (c) Acetal

g statements about alcohols

prepared by the reaction of

46. Which of the following

57. Ethers are generally

is correct?

(a). Secondary alcohols are oxidized to aldehydes. alcohols with:

(b). Tertiary alcohols are easily oxidized to carboxylic acids (a) Alkali metals (b) Carboxylic acids (c) Alkyl acids. halides (d) Acyl chlorides

(c) Primary alcohols are oxidized to carboxylic acids

under strong oxidation conditions. 58. In Williamson ether synthesis. the reaction

(d). Alcohols cannot form hydrogen bonds. typically involves an alkoxide ion and a:

F (a) Alkene (b) Alkyl halide (c) Ketone (4)

s fastest with Carboxylic acid

47. Which of the following alcohols react

Lucas reagent (HCl/ZnCl₂)?

(a). Methanol (b) Ethanol (c) 2-Butanol (4) 2- 59, Which:

amine? (a) CH₃

Methyl-2-propanol

h of the following is an example of a primary

-NH-CH₃

(b) CH₃CH₂-

CH₃

60. Which is

compound CH₃CH₂NH₂?

(a) Ethylamine

Ethylamino

OPTION A

NH₂ (c) (CH₃)₃N (4) CH₃CH₂CH₂-NH-

the correct IUPAC name for the

(b) Aminoethane (c) Ethanamine (d)

61. The dipole-dipole interaction in aldehyde and

ketones is due to
 molecular weight (b) the presence of C-C
 (a) their relatively high
 sigma bonds
 (c) the presence of carbonyl bond (d) their similarities
 in their molecular structure

62. Between alkanals and alkanones of similar
 molecular weights, W
 alkanals are more reactive because 0
 hich is more reactive and why? (a)
 f the presence of
 hydrogen bond (b) ketones are more reactive because
 of the presence 0
 f hydrogen bond (c) alkanals are more
 reactive because of the absence of hydrogen bond (d)
 ketones are more reactive becaus'
 hydrogen bond

63. Which reagent is used t
 ketones to alcohols? (a) BHs (6) NaBHa
 CrOs

64, Which reaction can be
 an aldehyde and a ketone
 Oxidation reaction (Cc) Reduction
 Halogenation

End of paper
 e of the absence of

© reduce aldehydes and
 (c) KMnOs (d)

used to distinguish between
 2 (a) Grignard reaction (b)
 reaction _(d)

65. What is the TUP
 from

Propionitrile

Methyl

66. WI

'synthesizing 1

Reduction of a

with sodium cy

67, Nitriles can be hydroly

following compoun

AC name for the nitrile derived

(a) Propanenitrile (b)

(d) ies

propanoic acid?

(c) Propan-| -nitrile

propanenitrile

s a common method for

itriles? (a) Hydrolysis of amides (b)

|dehydes (c) Reaction of alkyl halides

anide (d) Dehydration of alcohols

hich of the following i

zed to form which of the

ds? (a) Alcohols (b) Carboxylic

acids (c) Ketones (d) Aldehydes

68. Which reagent is comm

aldehydes to carboxylic acids? (

(b) Potassium permangana

only used to oxidize

a) Sodium borohydride

te (c) Lithium aluminum

hydride (d) Hydrogen gas

69 Markownikoff's rule deal with the reaction of
unsymmetrical reagent with ----- alkene.

(a) linear (b) unsymmetrical (c)

symmetrical (4) cyclic

70.

carboxylic acid

What is the general functional group of

5? (a) -C=O (b) -COOH (c) -CHO

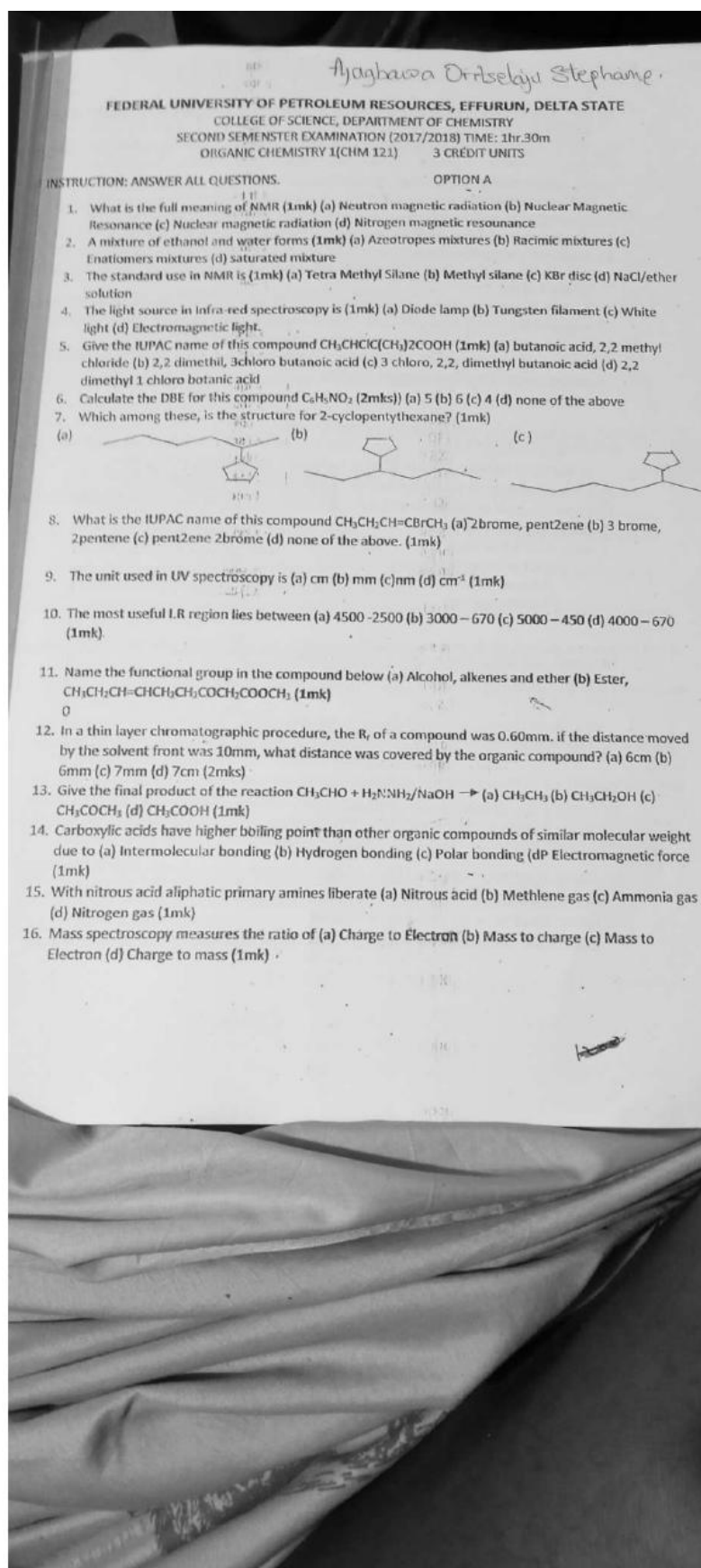
(d) -C=N

j

Paper 2 of 2 — CHM 121

Level 100 • Second Semester • PastQuestion

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17. The principle of chromatography is (a) Absorption (b) Polarity (c) Density (d) Charges on a peak (1mk)

18. What is base peak in Mass spectroscopy (1mk) (a) Last peak (b) Tallest peak (c) First Peak (d) peak

19. Kjeldahl's method associated with determination of which element (1mk) (a) Bromine (b) Nitrogen (c) Chlorine

20. Give two examples of ortho/para directors (a) OR/NO (b) OH/NH₂ (c) NR₂/CO (d) CN/NO₂

21. Give two examples of meta directors (a) NO₂/CHO (b) CO/OH (c) R/NO (d) NHR/O (1mk)

22. Elemental analysis of a compound was 85.8% C and 14.3% H. 0.031g of this compound at 760 mmHG, occupies 110ml. Calculate the molecular formula (2mk) (a) C₅H₁₂ (b) C₈H₁₂Cl₂ (d) None of the above

23. 6.23mg of a compound on combustion produced 9.17mg of CO₂ and 3.78mg of water. 352 vapour of the compound at 27°C and 750mmHG weighed 0.890g. Determine the empirical molecular formula of the compound. (2mk). (a) CH₂O/C₂H₄O₂ (b) C₈H₁₇CHO/C₆H₁₀

24. Which reagent is required for oxymercuration-demercuration reaction sequence during an alcohol as shown in the equation below: (1mk)

$$\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}=\text{CH}_2 \xrightarrow{\text{Which Reagent?}} \text{CH}_3\text{CH}(\text{CH}_3)\text{CH}(\text{OH})\text{CH}_3$$

(a) NaOH/H₂O (b) KMnO₄/H₂O (c) Hg(OAc)₂/H₂O and NaBH₄ (d) BH₃ in H₂O₂/HO

25. What element do all organic compounds contain? (1mk) (a) H (b) C (c) Fe (d) O

26. To what organic family do this molecule belong? H-C≡C-H? (a) alcohol (b) alkene (c) carboxylic acid (1mk)

27. What is the IUPAC name of this compound? (1mk)

(a) 01,3-pentamethylpropane (b) 1,1,3,3,3-Tetramethylbutane (c) 2,4,4-trimethylpentane (d) 2,2,4-Trimethylpentane

$$\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ | \quad | \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH} \\ | \quad | \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$$

28. What is the name of the compound below? (1mk) (a) 3-methylenehexane (b) 2-propyl-4-ethyl-4-pentane (c) 2-ethyl-1-pentene

$$\begin{array}{c} \text{H} \quad \text{H} \\ | \quad | \\ \text{CH}_2=\text{CH}-\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \\ | \\ \text{CH}_2\text{CH}_2\text{CH}_3 \end{array}$$

29. What is the name of the major product from this reaction? (1mk)

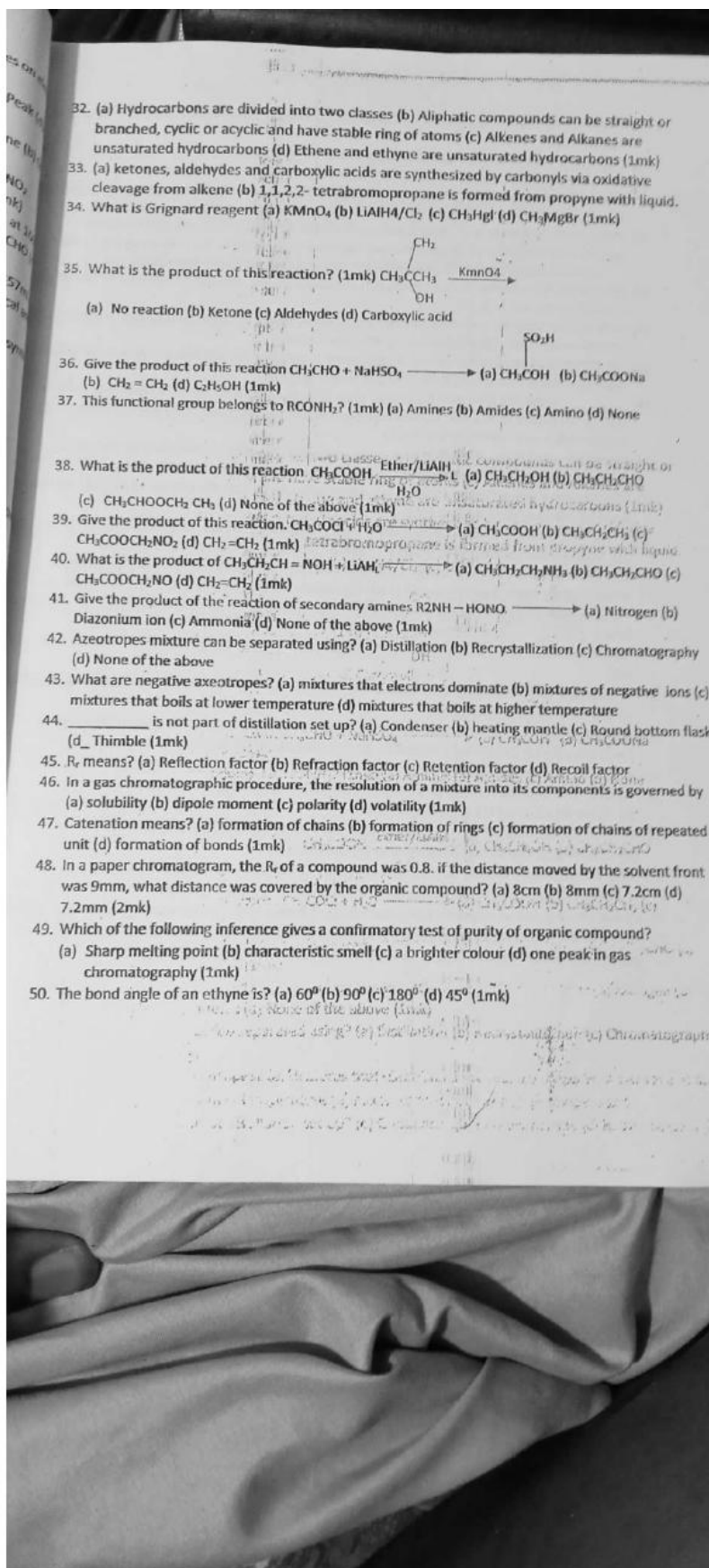
$$\text{CH}_3-\text{C}\equiv\text{C}-\text{H} \xrightarrow[\text{excess}]{\text{HCl}}$$

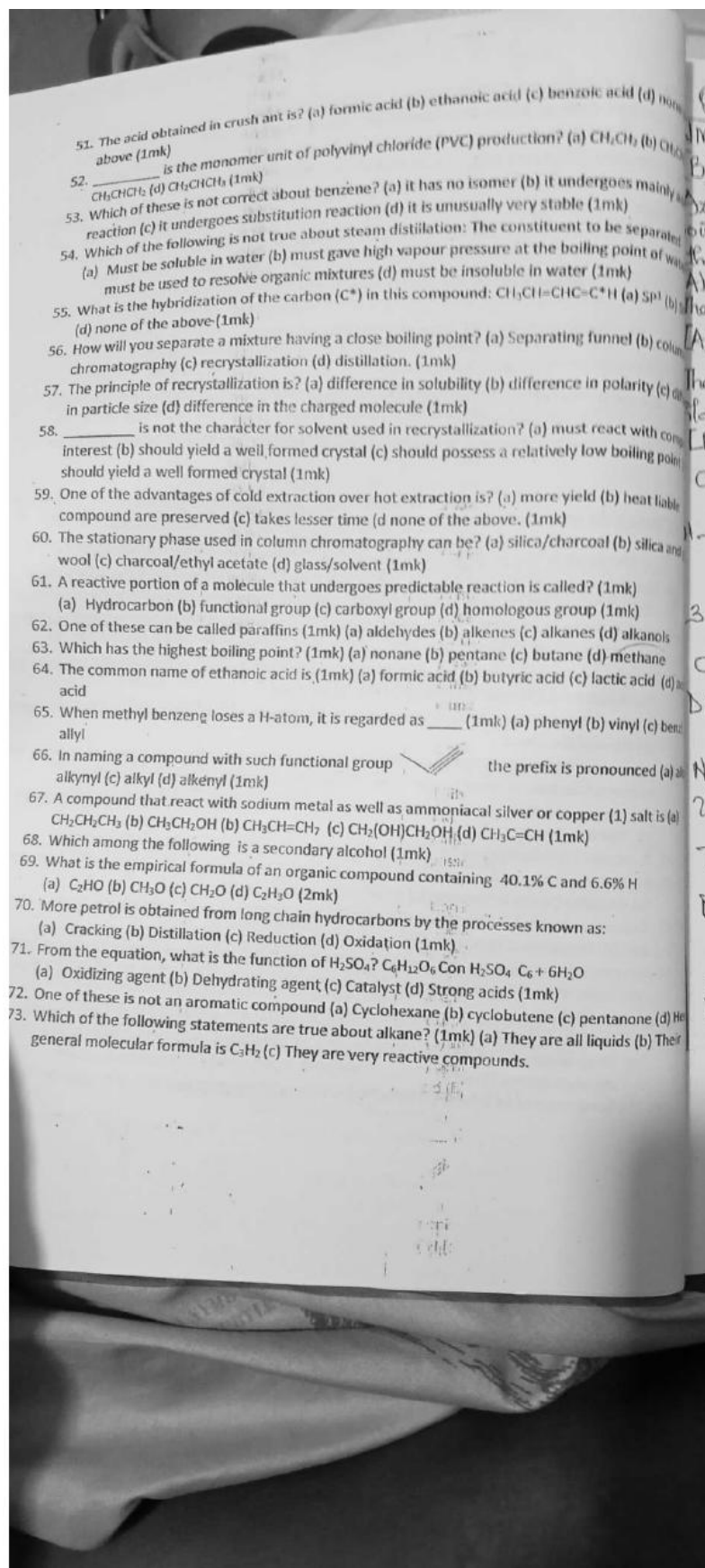
(a) CH₃-CCl=CH₂ (b) CH₃-CH=CHCl (c) CH₃-CHCl₂ (d) CH₃-CH₂-CH₂-CH₃

30. Which is 2,5-dimethylhexane? (1mk) (a) CH₃-CH(CH₃)-CH₂-CH(CH₃)-CH₂-CH₃ (b) CH₃-C(CH₃)₂-CH₂-CH₂-CH₂-CH₃ (c) CH₃-CH(CH₃)-CH(CH₃)-CH₂-CH₂-CH₃ (d) CH₃-CH(CH₃)-CH₂-CH₂-CH(CH₃)-CH₃

Aliphatic hydrocarbons are (1mk) (a) Hydrocarbons with aromatic rings (b) Nonane, dicyclopentadiene, cyclohexene (c) Cyclohexane, 1,3cyclopentadiene, 1,3,5,7-cyclooctatetraene (d) Benzene, n-heptane

Choose the odd one out of option A – D for question 32 and 33





ORGANIC CHEMISTRY [2017/2018] (22)

NMR → Nuclear Magnetic Resonance

(B)

Azeotropic mixtures are mixtures with constant boiling points. Examples: Nitric acid and water, water and ethanol, water and benzene etc.

(A)

The standard used in NMR is tetramethyl silane.

[A]

The light source in infra red spectroscopy is tungsten filament.

[B]

CH3CH(Cl)C(CH3)2COOH

3-chloro, 2,2-dimethylbutanoic acid.

C6H5NO2

DBE = $\frac{[\text{No. of C atoms} \times 2] + 2}{2} + [\text{No. of trivalent atoms}] + \frac{[\text{No. of univalent atoms}]}{2}$

No. of C atoms = 6

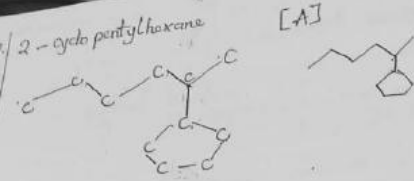
Univalent atoms = 5

Trivalent atom = 1

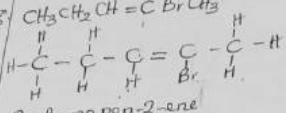
DBE = $\frac{[(6 \times 2) + 2]}{2} + 1 - 5 = \frac{14 + 2}{2} - 5 = \frac{16}{2} - 5 = 8 - 5 = 3$

[A]

7. 2-cyclopentylhexane [A]



8. $\text{CH}_3\text{CH}_2\text{CH}=\text{C}(\text{Br})\text{CH}_3$ [A]



2-bromopent-2-ene

9. The unit used in IR spectroscopy is nanometres (nm) [C]

10. The most useful I.R. region lies between 4000-670 [B]

11. $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}_2\text{COOCH}_2\text{CH}_2\text{COOCH}_3$
Ester, Alkanone, Alkene [B]

12. $R_f = \frac{\text{Distance moved by the compound}}{\text{Distance moved by the solvent}}$ [B]

$0.6 \text{ mm} = \frac{\text{Distance moved by the compound}}{10 \text{ mm}}$

Distance moved by the compound = $0.6 \times 10 = 6 \text{ mm}$.

13. $\text{CH}_3\text{CHO} + \text{H}_2 \xrightarrow{\text{Ni}/\text{NaOH}} \text{CH}_3\text{CH}_2\text{OH}$

Carboxylic acids have higher boiling point than other organic compounds of similar molecular weight due to hydrogen bonding.

[6]

Nitrogen gas [6]

Mass spectroscopy measures the ratio of mass to charge

[6]

The principle of chromatography is based on polarity.

B

Nitrogen [6]

B

Find the Empirical Formulae

$$\begin{array}{rcl} \text{C} & : & \text{H} \\ \hline 85.8 & : & 14.3 \\ 12 & & 1 \\ \hline 7.15 & : & 14.3 \\ 7.15 & & 7.15 \\ 1 & : & 2 \end{array}$$

Empirical formulae $\Rightarrow \text{CH}_2$

$m = 0.30 \text{ g}$
 $T_1 = 100^\circ\text{C} = 100 + 273 = 373 \text{ K}$
 $P_1 = 760 \text{ mmHg}$
 $V_1 = 110 \text{ mL}$
 At STP
 $P_2 = 760 \text{ mmHg}$
 $V_2 = ?$
 $T_2 = 0^\circ\text{C} = 0 + 273 = 273 \text{ K}$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 = \frac{P_1 V_1 T_2}{P_2 T_1}$$

$$V_2 = \frac{760 \times 110 \times 273}{760 \times 373}$$

$$= 80.5 \text{ mL}$$

~~number of moles =~~
 $\text{molar mass} = \frac{m \times V \text{ at STP}}{V_2}$
 $= \frac{0.30 \times 22400}{80.5}$
 $= 83.76 \text{ g/mol}$

Molecular formulae = (Empirical formula) n
 $n(\text{CH}_2) = 83.76$
 $n[12 + (2 \times 1)] = 83.76$
 $n[14] = 83.76$
 $n = \frac{83.76}{14} \approx 6$

$\text{Molecular Formula} = n[\text{CH}_2] = \text{C}_6\text{H}_{12} = \text{C}_6\text{H}_{12}$
 $[\text{H}]$

If 6.23g of organic compound gives 9.71mg CO_2 and 3.76mg H_2O

$\text{C}_x\text{H}_y + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

% Composition of C in CO_2
 $= \frac{12}{12 + (16 \times 2)} \times \frac{100}{100} = 40.14\%$

% Composition of H in H_2O
 $= \frac{2}{2 + (16 \times 1)} \times \frac{100}{100} = 11.11\%$

$\therefore \% \text{C} = 100 - [40.14 + 11.11] = 53.12\%$

$\text{C} : \text{H} : \text{O}$
 $\frac{40.14}{12} : \frac{53.12}{16} : \frac{53.12}{16}$
 $3.35 : 3.32 : 3.32$
 $1 : 2 : 1$

Empirical formula = CH_2O

Calculate volume at STP

$$V_2 = \frac{P_1 V_1 T_2}{P_2 T_1}$$

$$V_2 = \frac{750 \times 357 \times 273}{760 \times 300} \rightarrow 320.6 \text{ ml}$$

If 320.6 ml gases = 0.89 g
 1 ml at STP = x g

$$x = \frac{0.89 \times 22400}{320.6} = 62.18 \text{ g}$$

Molecular Formulae = (Empirical formulae) n = MM

$$\Rightarrow n(\text{CH}_2\text{O}) = 62.18$$

$$\Rightarrow n(12 + (2 \times 1) + 16) = 62.18$$

$$\Rightarrow 30n = 62.18$$

$$\Rightarrow n = \frac{62.18}{30} = 2$$

Molecular Formula = $n(\text{CH}_2\text{O})$

$$= 2(\text{CH}_2\text{O})$$

$$= \text{C}_2\text{H}_4\text{O}_2$$

 [A]

24 B

25 [B], Carbon

26 None of the above

$$V_2 = \frac{P_1 V_1 T_2}{P_2 T_1}$$

$$V_2 = \frac{760 \times 357 \times 273}{760 \times 300} = 320 \text{ mL}$$

If 320 mL gas = 0.89 g
 1 mL at STP = x g

$$x = \frac{0.89 \times 22.400}{320} = 62.18 \text{ g}$$

Molecular formulae = (Empirical formula) n = MM

$$\Rightarrow n(\text{CH}_2\text{O}) = 62.18$$

$$\Rightarrow n(12 + (2 \times 1) + 16) = 62.18$$

$$\Rightarrow 30n = 62.18$$

$$\Rightarrow n = \frac{62.18}{30} = 2$$

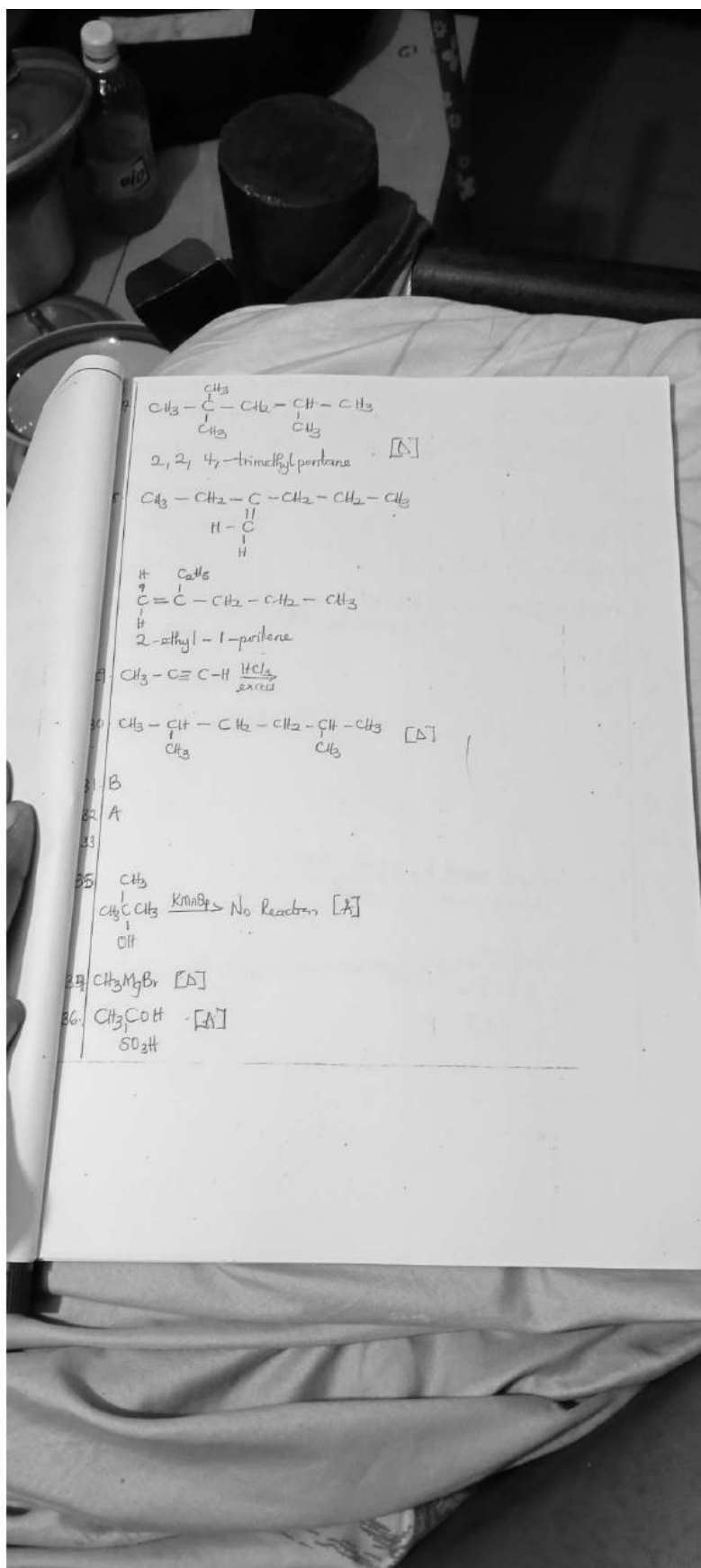
Molecular formula = $n(\text{CH}_2\text{O})$

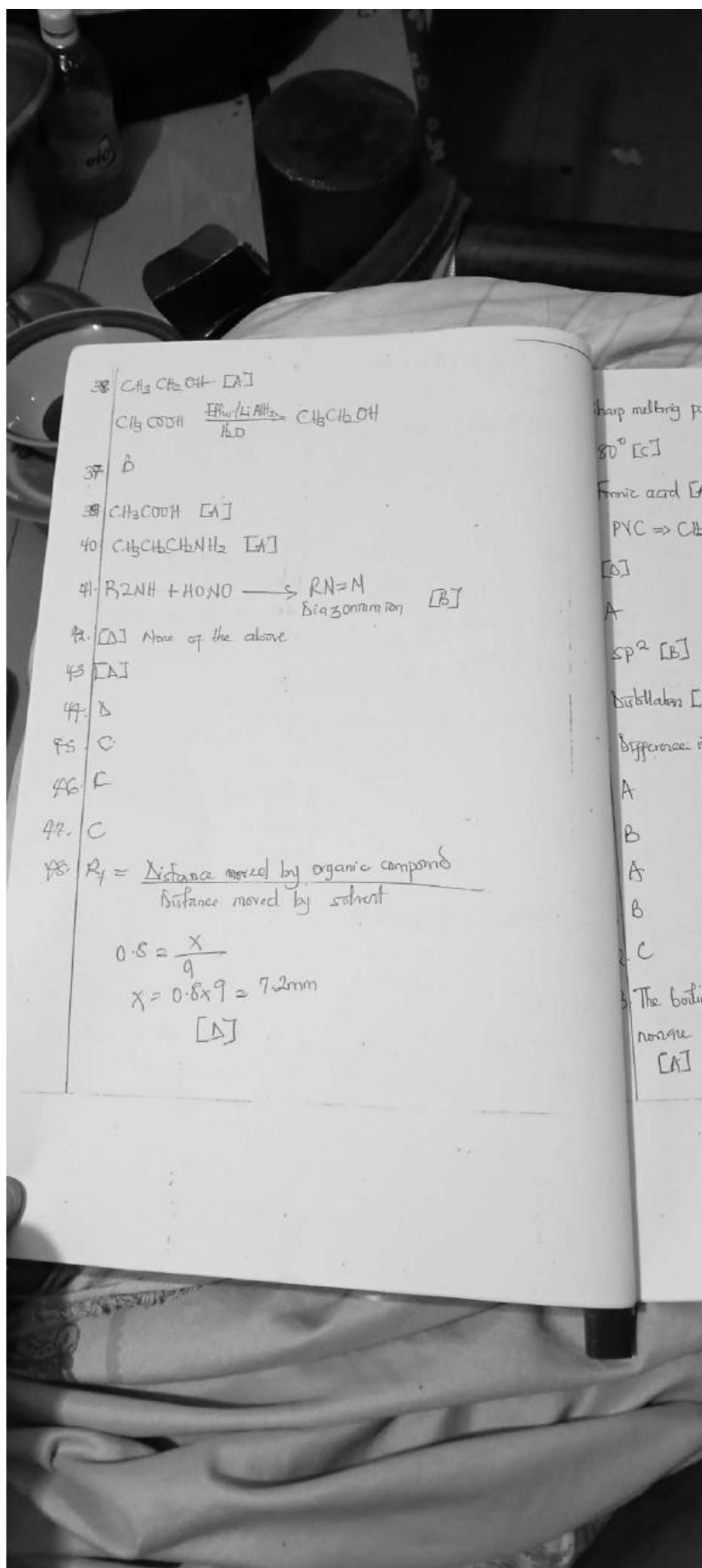
$$= 2(\text{CH}_2\text{O})$$

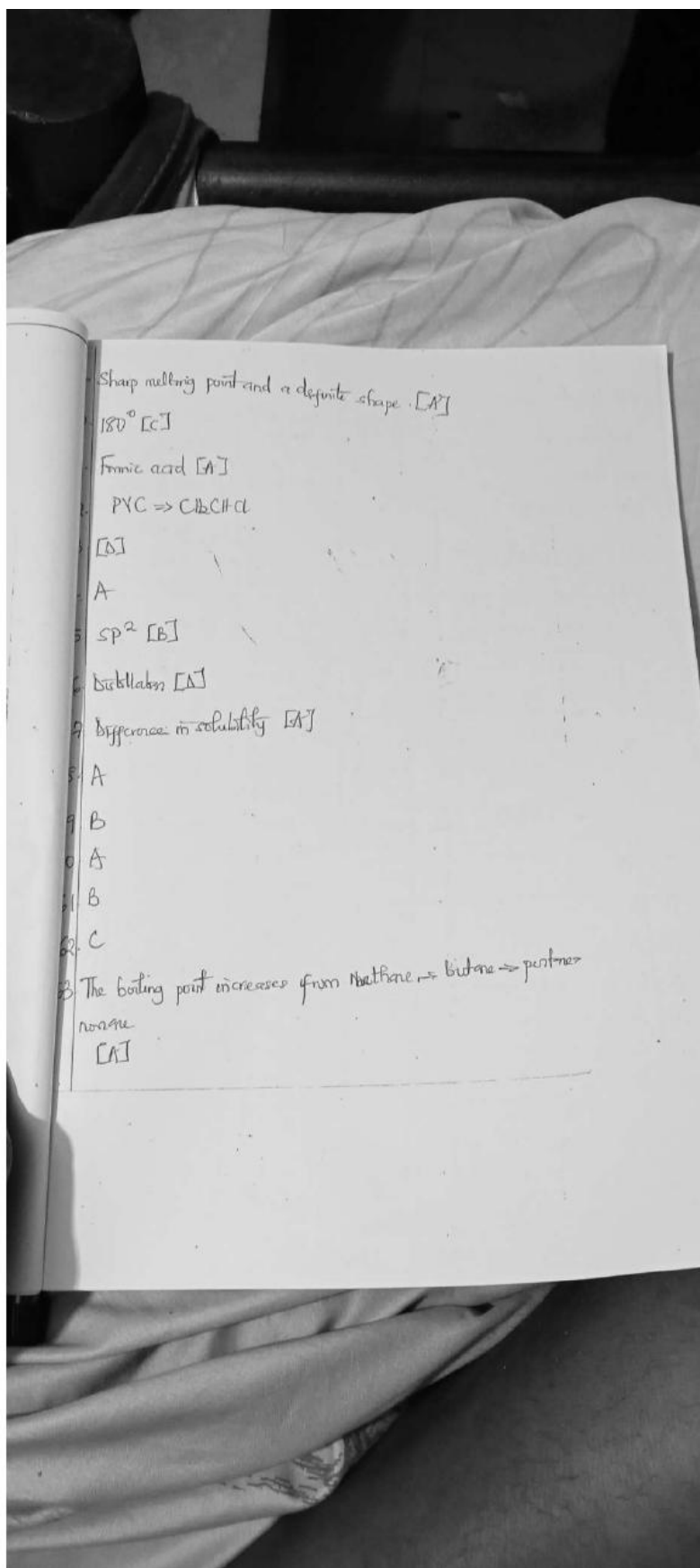
$$= \text{C}_2\text{H}_4\text{O}_2$$

 [A]

24. B
 25. B, Carbon
 26. None of the above







64. Acetic Acid Δ

65. Benzyl.

66. Alkyne

67. $\text{CH}_3\text{C}\equiv\text{CH}$ 68. $\text{CH}_3\text{CH}(\text{COOH})\text{CH}_3$

$$\begin{array}{rcl}
 \text{C} & : & \text{H} : \text{O} \\
 40.1 & : & 6.6 : 53.3 \\
 \hline
 12 & & 1 \quad 16 \\
 3.34 & : & 6.6 : 3.33 \\
 \hline
 3.33 & & 3.33 \quad 3.33 \\
 & & 1 \quad 2 \quad 1
 \end{array}$$

 $\Rightarrow \text{CH}_2\text{O} \quad [\text{C}]$

70. A

71. Δ

72. A

73. Δ 74. Δ

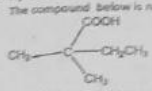
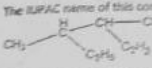
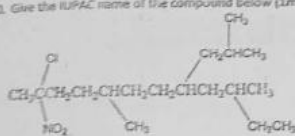
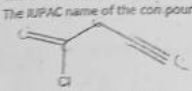
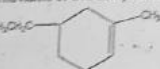
75. C

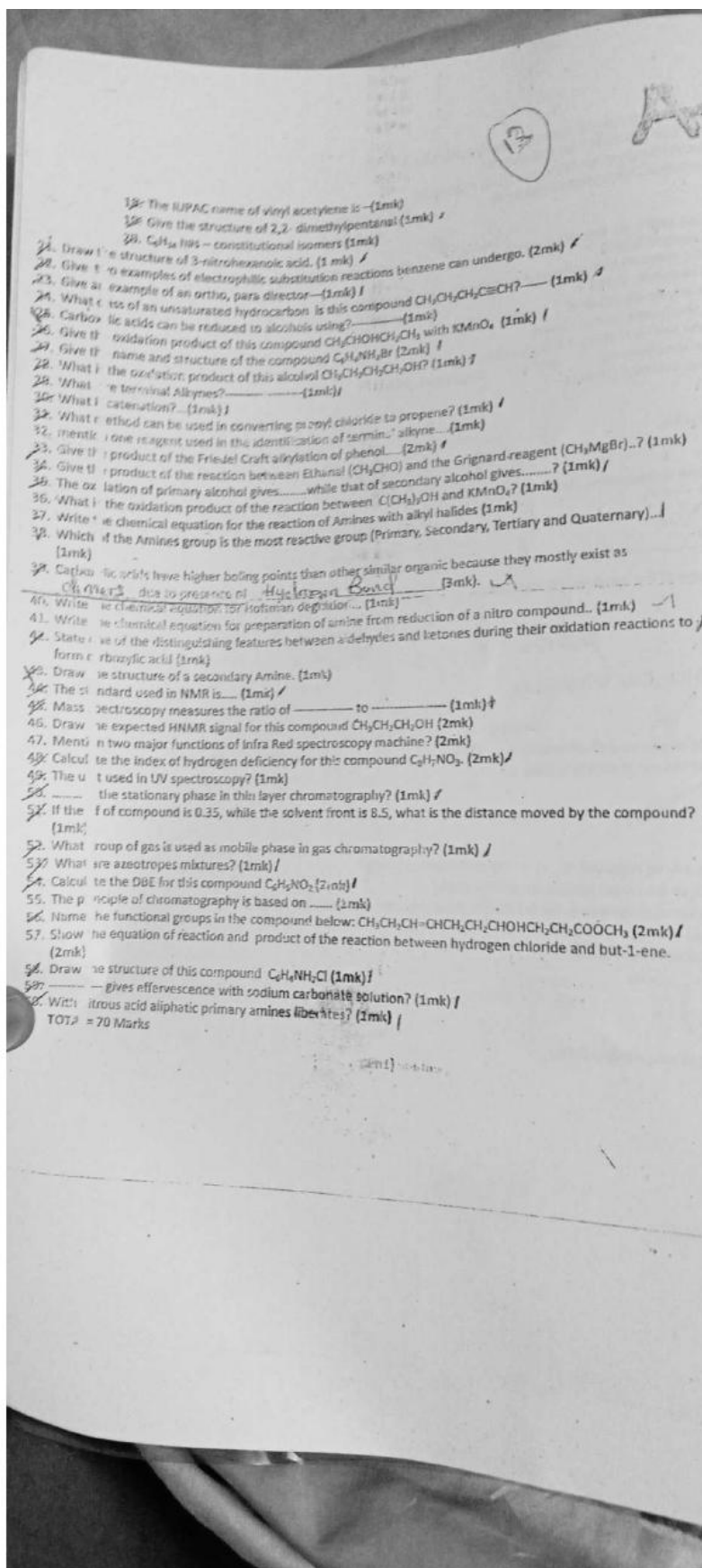
	GENERAL MOLECULAR FORMULA	FUNCTIONAL GROUP	PREFIX	SUFFIX
alkanes (halides)	$R = C_nH_{2n+1}$ $R-X$	$-X$ halogen		-ane or ide
alkane	C_nH_{2n+2}	$-C-C-$		-ane
alkene	C_nH_{2n}	$C=C$		-ene
alkyne	C_nH_{2n-2}	$C \equiv C$		-yne
oxygens (ethers)	$R-O-R'$	$-O-$ ether linkage	alkoxy	-ane or -ether
primary amine	$R-NH_2$	$-NH_2$	Amino-	-amine or -ane
secondary amine	$R-\overset{\overset{H}{ }}{N}-R'$	$>NH$ N-substituted	Amino-	-amine or -ane
tertiary amine	$R-\overset{\overset{R''}{ }}{N}-R'$	$>N:$ N,N-substituted	Amino-	-amine or -ane
alcohols	$R-O-H$	$-OH$	Hydroxy	-ol
ketones (ones)	$R-\overset{\overset{O}{ }}{C}-R'$	$C=O$ Carbonyl group	-oxo- or keto	-one
aldehydes (als)	$R-\overset{\overset{O}{ }}{C}-H$	$-\overset{\overset{O}{ }}{C}-H$	Alko or formyl	-al
carboxylic acids (oic acids)	$R-\overset{\overset{O}{ }}{C}-O-H$	$-\overset{\overset{O}{ }}{C}-O-H$	Carboxy	-oic acid.

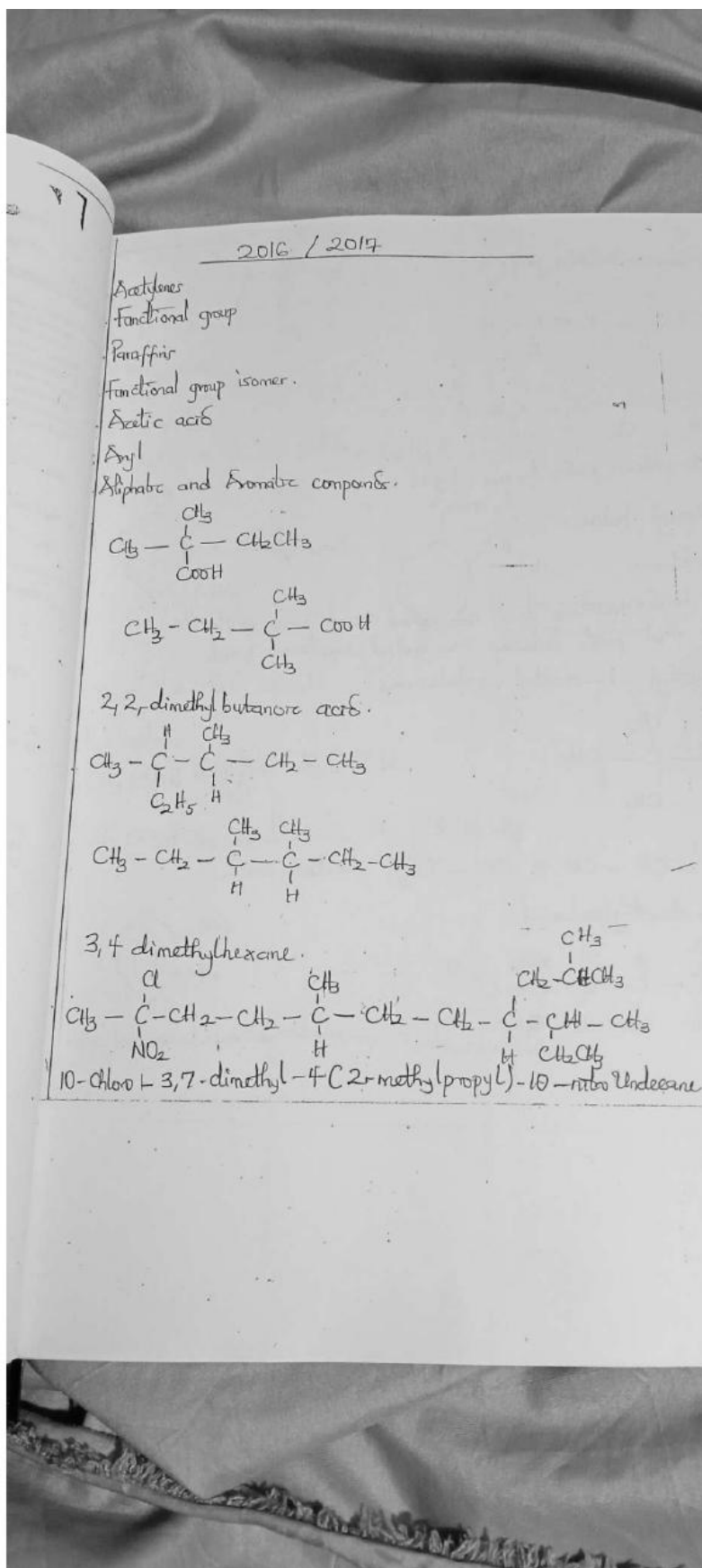
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
DEPARTMENT OF CHEMISTRY
2ND SEMESTER EXAMINATION 2016/2017 SESSION
GENERAL ORGANIC CHEMISTRY

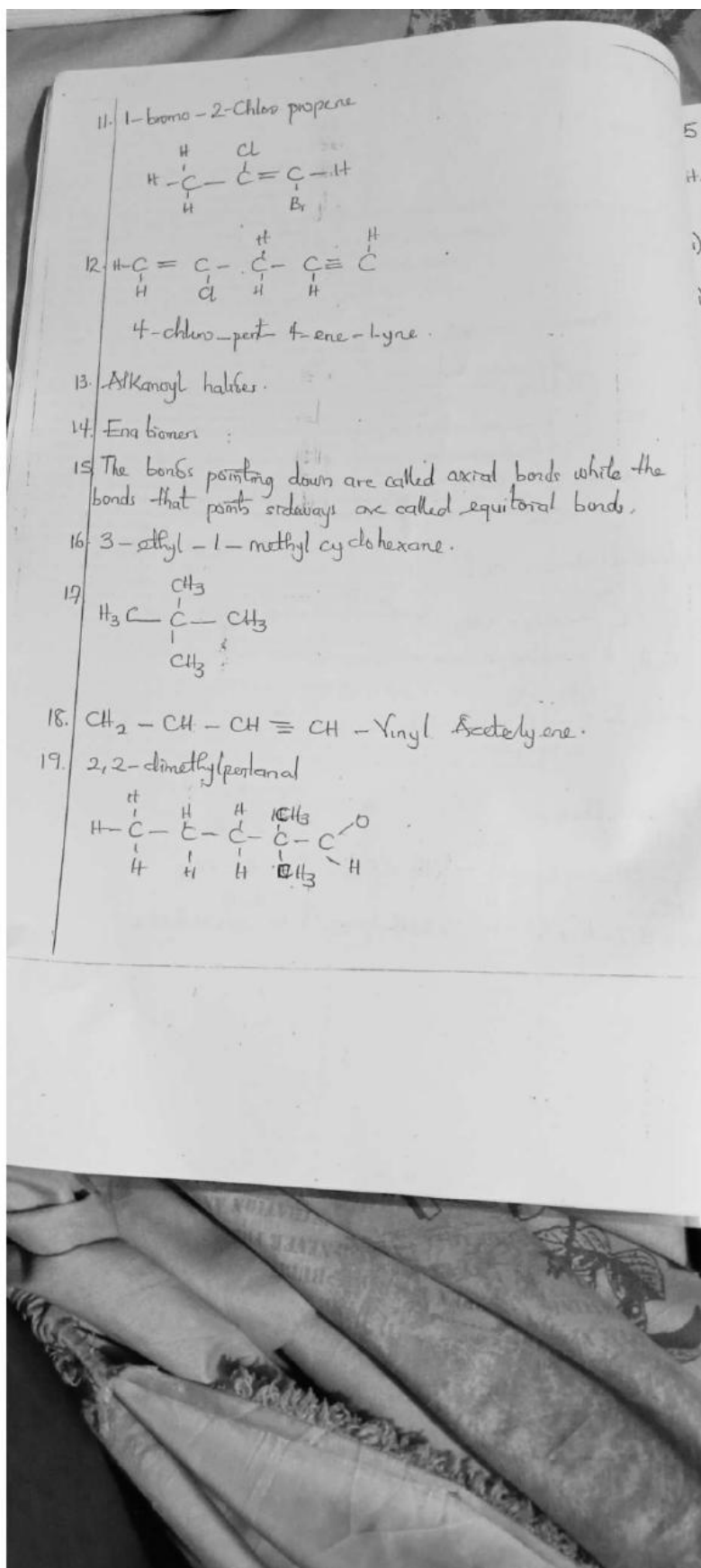
CHM 121
1HR 30MINS.
OPTION A

ANSWER ALL QUESTIONS

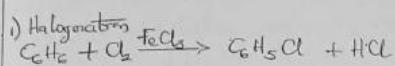
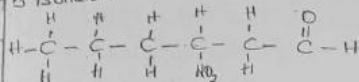
- In naming a compound with such functional group  what family of an organic compound does it belong to? (1mk) ✓
- A reactive portion of a molecule that undergoes predictable reaction is called- (1mk) ✓
- Alkanes can equally be called - (1mk) ✓
- Ethanol and Ethanal may be classified as - isomers (1mk) ✓
- The common name of ethanoic acid is - (1mk) ✓
- When methyl benzene loses a H-atom, it is regarded as - (1mk) ✓
- Hydrocarbons can be classified into - and - compounds (1mk) ✓
- The compound below is named (1mk) ✓

- The IUPAC name of this compound is- (1mk) ✓ *Petroleum*

- Give the IUPAC name of the compound below (1mk) ✓

- The structure of (E)-1-bromo-2-chloropropene is - (1mk) ✓
- The IUPAC name of the compound is - (1mk) ✓

- Carbamoyl can equally be regarded as - functional group (1mk) ✓
- Optical isomers which are mirror images are called - (1mk) ✓
- In chair conformation of cyclohexane, the bonds that point sideways and down are called - and - bonds. (1mk) ✓
- The name of this compound is- (1mk) ✓

- Give the structure of neopentane (1mk) ✓



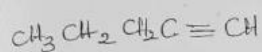
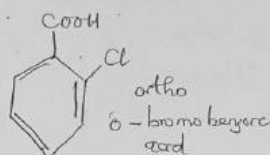
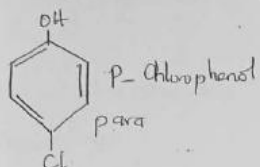
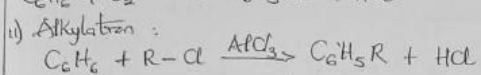




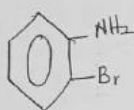
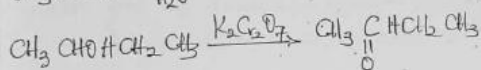
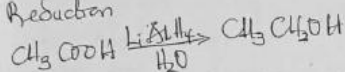
5 isomers:



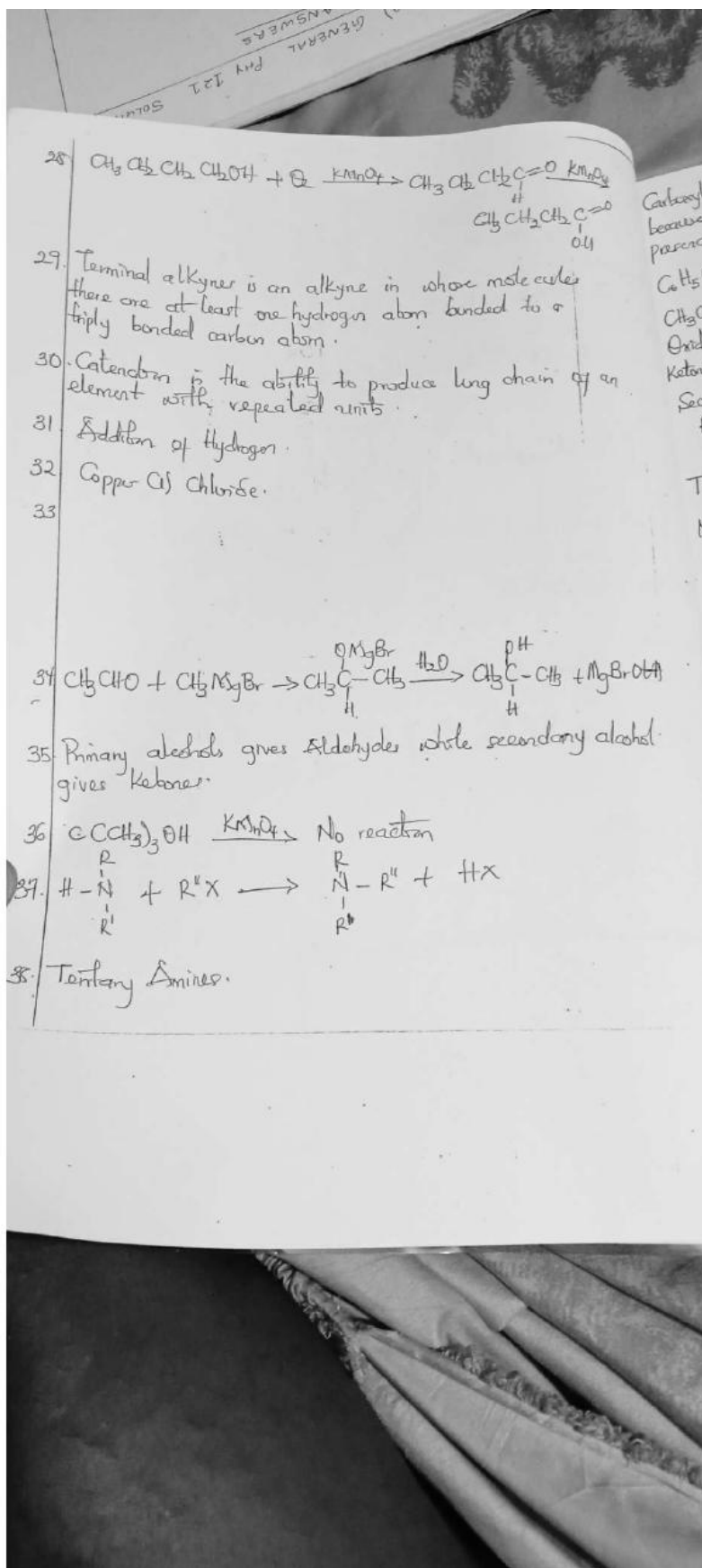
ii) Alkylation:

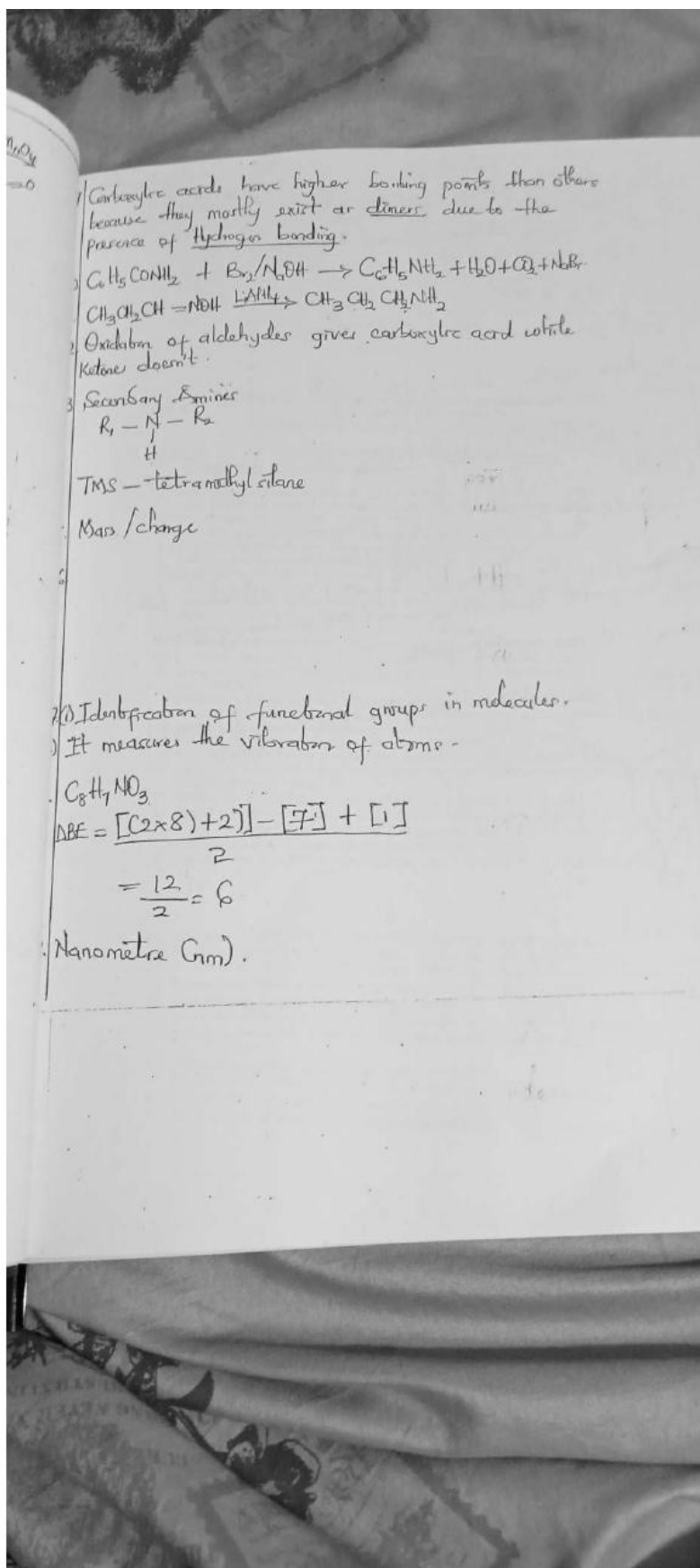


Reduction



~~2-bromo~~ 1-amino-2-bromo benzene 2-bromo aniline





51. $R_f = \frac{\text{Distance moved by component}}{\text{Distance moved by solvent}}$
 let X be the distance moved by solvent
 $0.35 = \frac{X}{8.5}$
 $X = 2.975$

52. Inert gas

53. Azeotropic mixtures are mixtures with constant boiling points.

54. $DBE = \frac{[(6 \times 2) + 2] - [5] + [1]}{2}$
 $C_6H_5NO_2$
 $= \frac{10}{2} = 5$

55. It's based on the difference in forces of attraction between components of a mixture and one stationary phase.

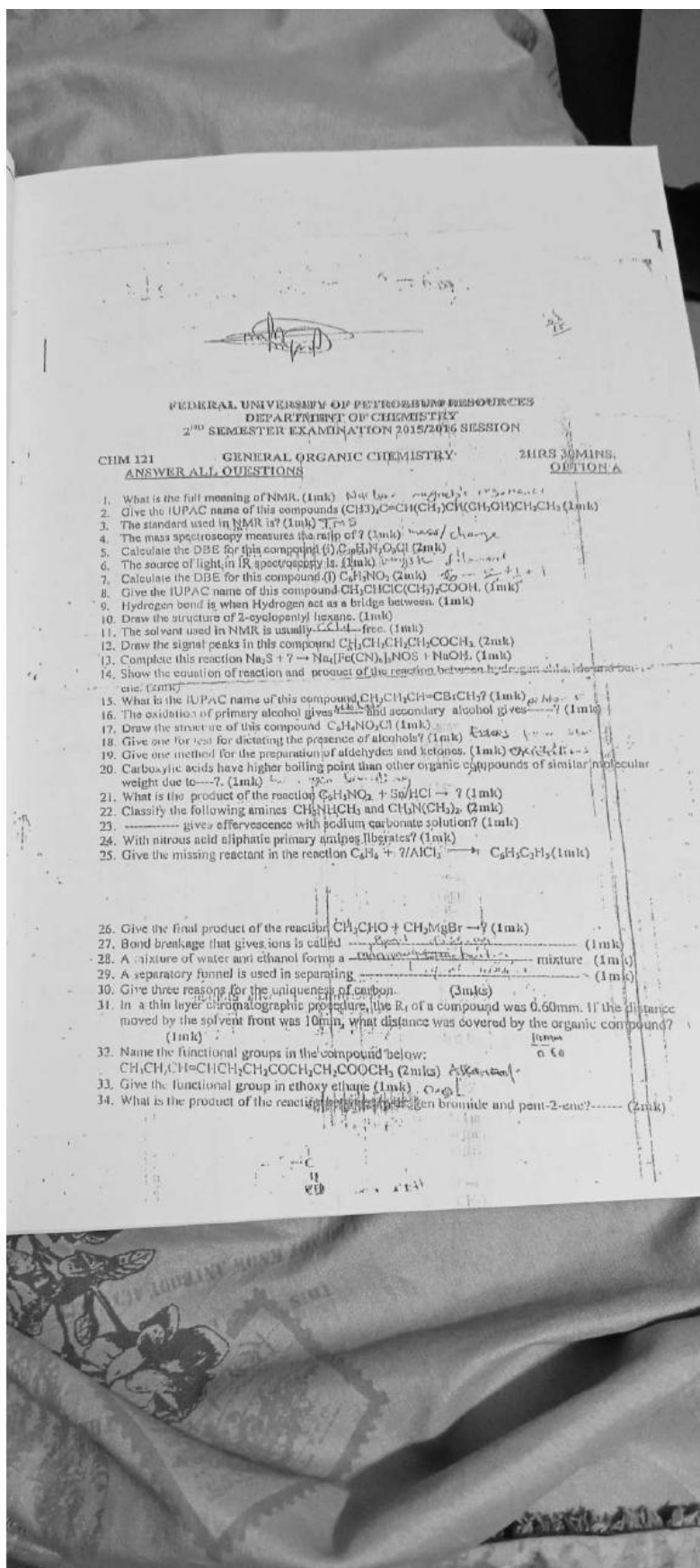
56. Alkenes, Alkanes, Alkanones

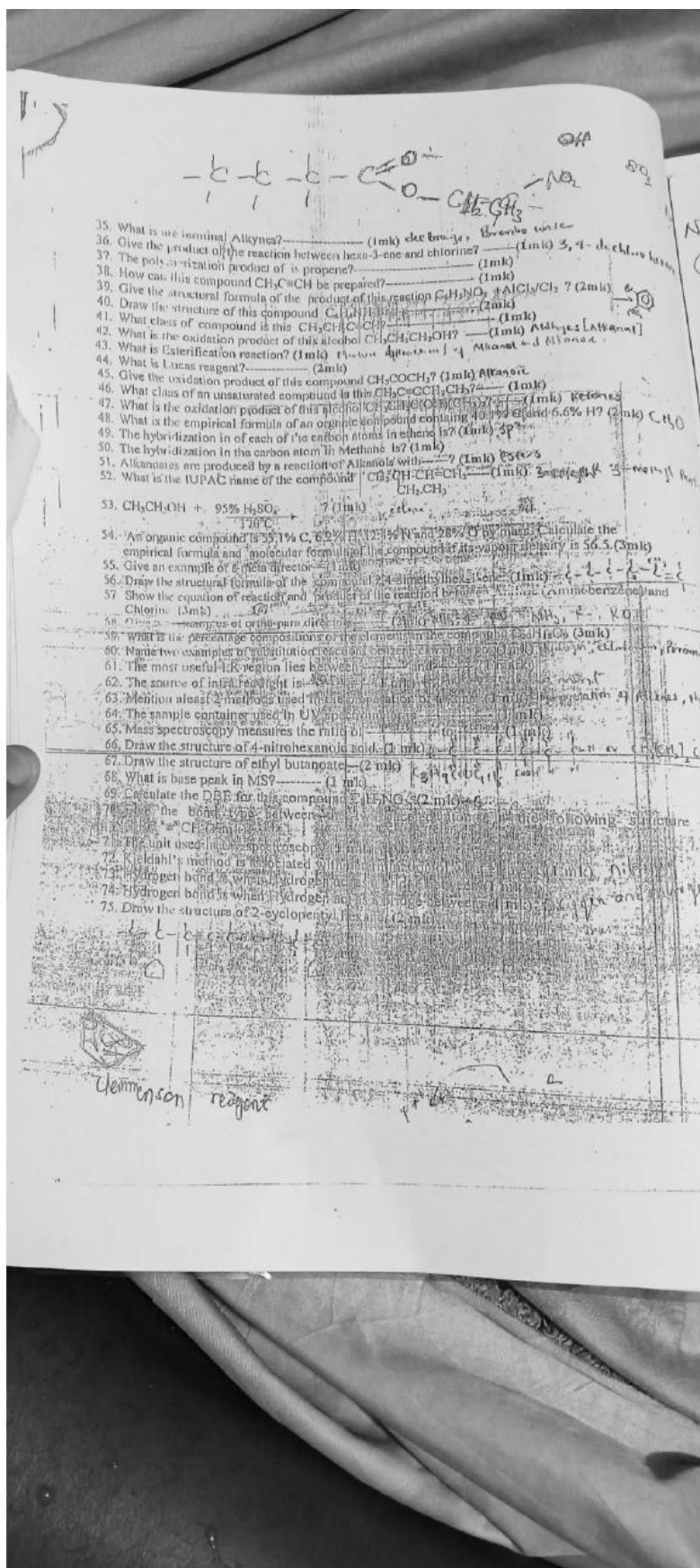
57. $CH_3CH=CH_2 + HCl \rightarrow CH_3CHClCH_3 + CH_3CH_2CH_2Cl$
 (Markov)

58. 

59. Ethanoic acid

60. Nitrogen





2015/2016

1. NMR - Nuclear Magnetic Resonance Spectroscopy.

2. $(CH_3)_2C = \overset{\overset{CH_3}{|}}{CH} \overset{\overset{CH_2OH}{|}}{C} CH_2CH_3$
 $CH_3 - C = \overset{\overset{H}{|}}{C} - CH_2 - CH_2 - CH_3$

3. TMS - Tetramethylsilane.

4. It measures the ratio of mass/charge.

5. $C_{10}H_8Na_2O_2Cl$
 $DBE = \frac{[No \text{ of Carbon atoms} \times 2] + 2 - [No \text{ of univalent atoms}] + [No \text{ of trivalent atoms}]}{2}$
 $= \frac{[10 \times 2] + 2 - [8 + 1] + [2]}{2}$
 $= \frac{18}{2} = 9$

6. Tungsten filament

7. $C_6H_5NO_2$
 $DBE = \frac{[No \text{ of Carbon atoms} \times 2] + 2 - [No \text{ of univalent atoms}] + [No \text{ of trivalent atoms}]}{2}$
 $= \frac{[6 \times 2] + 2 - [5] + [1]}{2}$
 $= \frac{10}{2} = 5$

8. $CH_3CH(Cl)C(CH_3)_2COOH$
 3-chloro-2,2-dimethylbutanoic acid.

9. Hydrogen acts as a bridge between two electronegative atoms.

10. $\text{CH}_3\text{CH}(\text{Cyclopentyl})\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

11. CCl_4

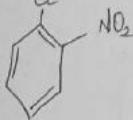
12.

13. $\text{Na}_2\text{S} + \text{Na}[\text{Fe}(\text{CN})_5\text{NO}] \rightarrow \text{Na}_4[\text{Fe}(\text{CN})_6]_5\text{NOS} + \text{NaOH}$
 purple complex
 Sodium-Turner solution mixed with sodium nitroprusside.

14. $\text{CH}_3\text{CH}=\text{CH}_2 + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{CH}_3\text{CH}(\text{Cl})\text{CH}_3$
 (minor)

15. $\text{CH}_3-\text{CH}_2-\text{CH}=\underset{\text{Br}}{\text{C}}-\text{CH}_3$
 2-bromopent-2-ene

16. The oxidation of primary alcohol gives Aldehyde and secondary alcohol gives Ketone.

17. $\text{C}_6\text{H}_4\text{NO}_2\text{Cl}$


18. i) Esters form sweet smelling esters when reacted with organic acids in the presence of dilute sulphuric acid.

ii) The presence of primary and secondary alcohols can be shown to be the ability to decolorize potassium permanganate solution or turn potassium or sodium dichromate solution to green from orange.

19. Oxidation

20. Hydrogen bonding

21. $C_6H_5NO_2 + Sn/HCl \rightarrow C_6H_5NH_2 + SnCl_2 + H_2O$

22. $CH_3-NH-CH_3$ - Secondary Amine.
 $CH_3N(CH_3)_2$ - Tertiary Amine.

23. Ethanoic acid.

24. Nitrogen

25. $C_6H_6 + C_2H_5Cl/AlCl_3 \rightarrow C_6H_5C_2H_5 + HCl$

26. $CH_3CHO + CH_3MgBr \rightarrow CH_3-\overset{\overset{O^+MgBr}{\parallel}}{C}-CH_3 \xrightarrow{H_2O} CH_3-\overset{\overset{OH}{\parallel}}{C}-CH_3 + MgBrOH$

27. Bond fission

28. Azeotropic Mixture.

29. Immiscible solvent phase of different densities.

30. It forms single, double and triple bonds with itself and other atoms.

(i) It forms long chains of repeated units (catenation) and various sizes of ring structures.

(ii) It has half-filled shell in the second period which confers some degree of stability.

31. Retention factor (R_f) = $\frac{\text{Distance moved by component}}{\text{Distance moved by solvent}}$

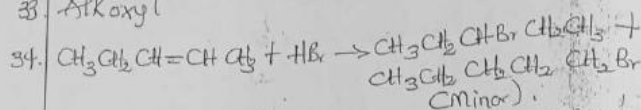
$$0.60 \text{ mm} \times \frac{\lambda}{10 \text{ mm}}$$

$$X = 0.60 \times 10$$

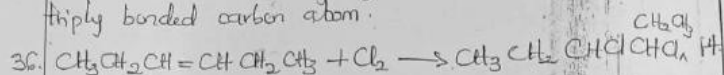
$$X = 6 \text{ mm}^2$$

32 Alkynes, Alkenones, Alkanates

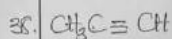
33. SPKoxyl



35. Terminal Alkyne is an alkyne in whose molecule there is at least one hydrogen atom bonded to a triply bonded carbon atom.

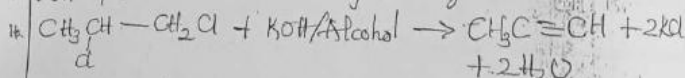


37. Isoprene

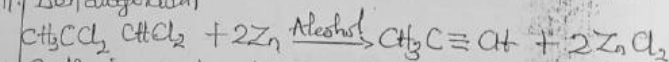


1. It can be prepared using the following

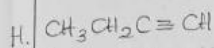
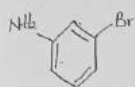
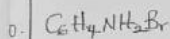
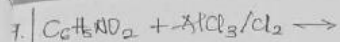
1. From 1,2-dihalides is dehydrohalogenation



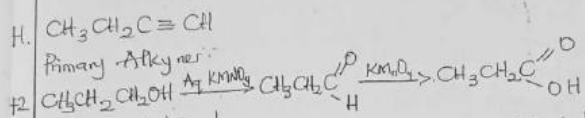
ii. Dehalogenation



iii. Synthesis of Non-terminal Alkynes

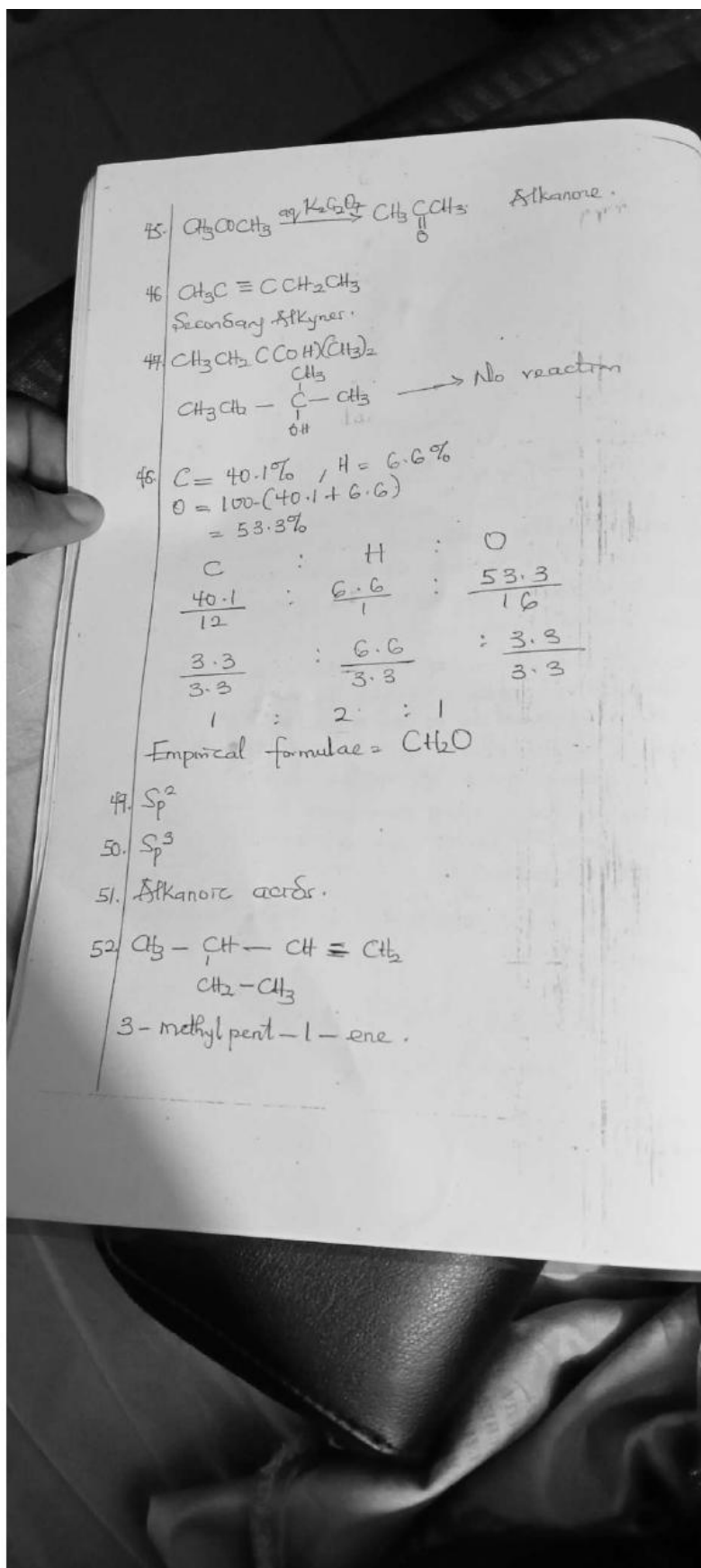


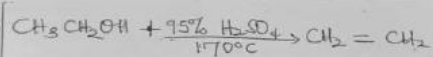
Primary Alkyne



13. It gives Butanol
Esterification Reaction is the reaction of alcohols with organic acids in the presence of mineral acids as catalyst in a reversible reaction to give esters and water.

14. Lucas reagent is a solution of anhydrous zinc chloride in concentrated hydrochloric acid. The solution is used to classify alcohols of low molecular weight. The reaction is a substitution in which the chloride replaces a hydroxyl group. A positive test is indicated by a change from clear and colourless to turbid signifying formation of a chloroalkane. The best results for this test is observed in tertiary alcohols as they form the respective alkyl halides fastest due to higher stability of the intermediate tertiary carbocation.





$$\begin{array}{cccc} \text{C} & : & \text{H} & : & \text{N} & : & \text{O} \\ \frac{53.1}{12} & : & \frac{6.2}{1} & : & \frac{12.4}{14} & : & \frac{2.8}{16} \\ \frac{4.4}{0.9} & : & \frac{6.2}{0.9} & : & \frac{0.9}{0.9} & : & \frac{1.8}{0.9} \\ 5 & : & 7 & : & 1 & : & 2 \end{array}$$

Empirical formulae = $\text{C}_5\text{H}_7\text{NO}_2$

Molecular formulae = $n(\text{Empirical formulae})$

$$56.5 \times 2 = n[(0 \times 5) + (1 \times 7) + (1 \times 14) + (2 \times 32)]$$

$$113 = n(7 + 14 + 64)$$

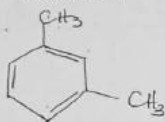
$$\frac{113}{85} = n \quad \therefore n = 0$$

$\therefore$ Molecular formulae = 0 ($\text{C}_5\text{H}_7\text{NO}_2$)

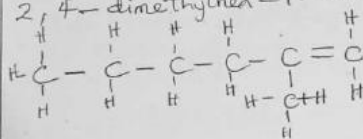
$$= 0$$

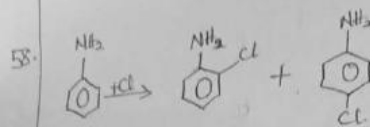
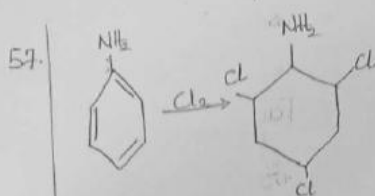
$\therefore \text{C}_5\text{H}_7\text{NO}_2$ has no molecular formulae.

5. Meta director.



6. 2,4-dimethylhex-1-ene





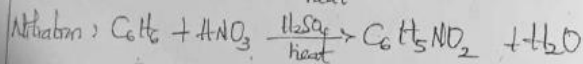
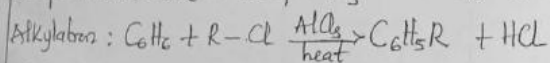
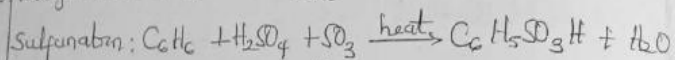
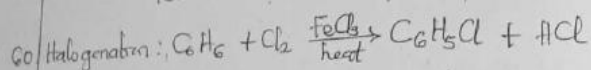
Examples of ortho-para director - Cl , Br , I , OH , NH_2

59. $\text{C}_{20}\text{H}_{18}\text{O}_4$
 $\% \text{ Composition} = \frac{[12 \times 20] + [18 \times 1] + [16 \times 4]}{402}$

$\% \text{ C} = \frac{240}{402} \times 100 = 59.7\%$

$\% \text{ H} = \frac{18}{402} \times 100 = 4.5\%$

$\% \text{ O} = \frac{144}{402} \times 100 = 35.8\%$

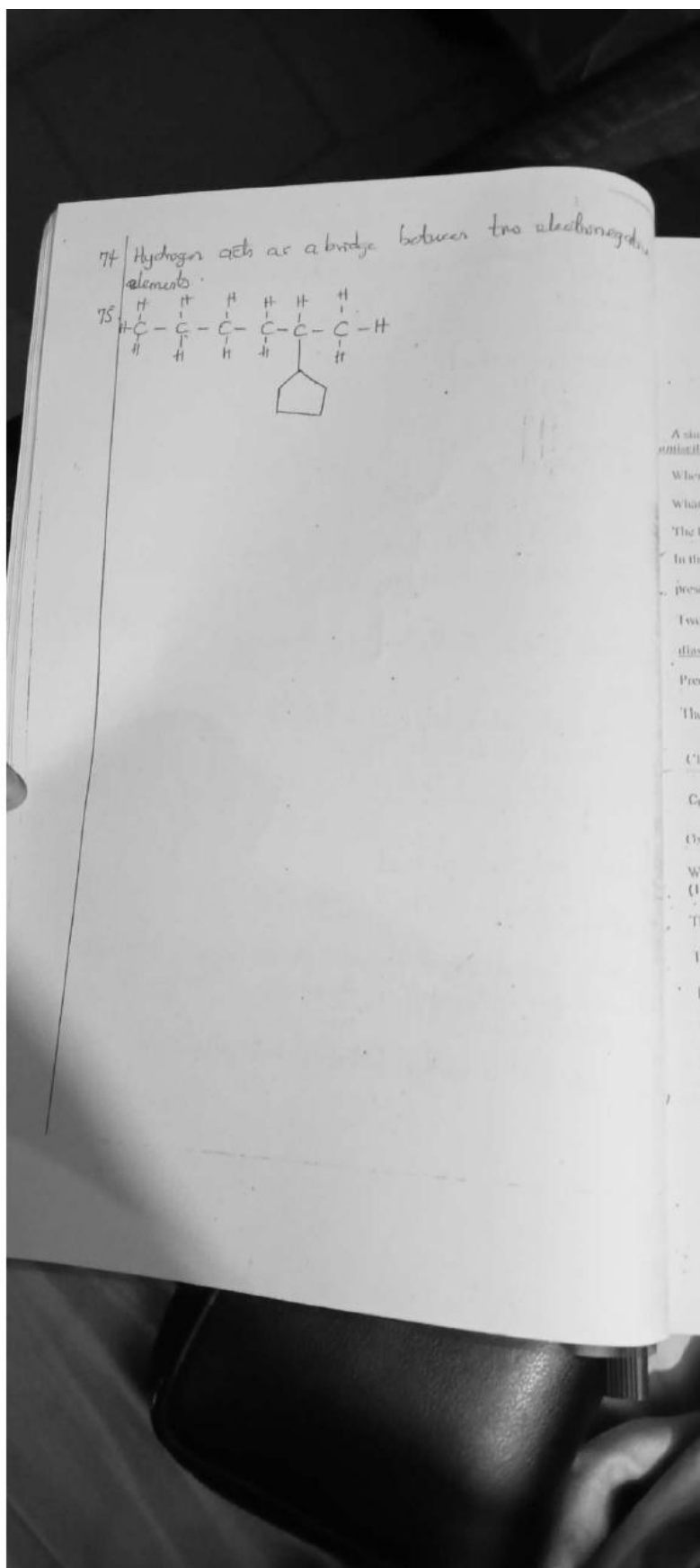


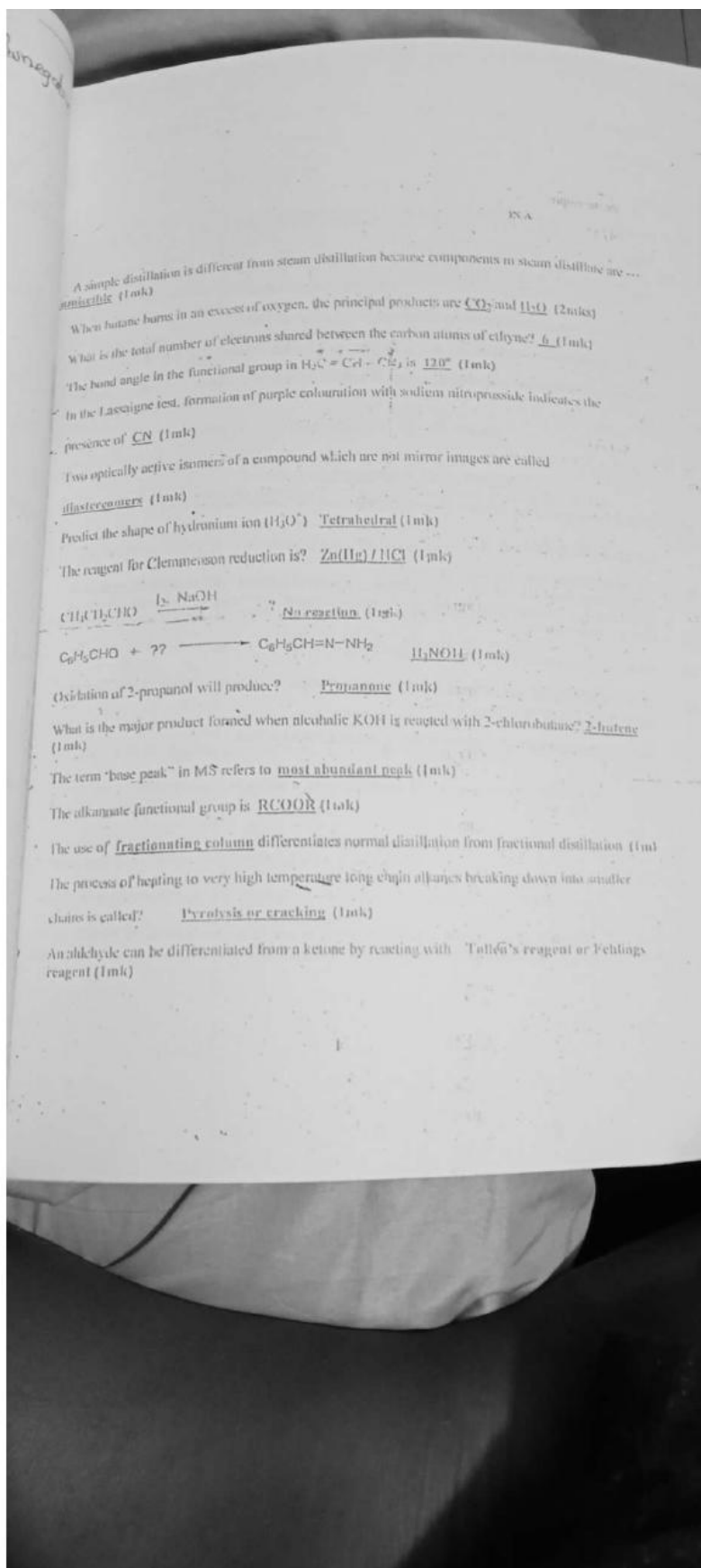
61. $4000 - 610\text{cm}^{-1}$

2. Tungsten Filament
3. i) Catalytic Hydrogenation of unsaturated hydrocarbons.
- ii) Alkyl Halides
- iii) Hantz reaction
- iv) Reducing Agent
- v) Kolbe's Electrolytic method.
4. Quartz
5. Mass/charge
6.
$$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{NO}_2 & & \text{H} & & \text{H} \\ & | & & | & & | & & | & & | \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{O} \\ & | & & | & & | & & | & & | & & \text{H} \end{array}$$
7. $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \underset{\text{O}}{\underset{||}{\text{C}}} - \text{O} - \text{CH}_2 - \text{CH}_3$
8. Base peak is the stable ion.
9. $\text{C}_8\text{H}_7\text{NO}_3$

$$\text{DBE} = \frac{[\text{No of C atoms} \times 2] + 2}{2} - \frac{[\text{No of univalent atoms}] + [\text{No of trivalent atoms}]}{2}$$

$$= \frac{[(8 \times 2) + 2]}{2} - \frac{[7 \times 1] + 1}{2} = \frac{12}{2} = 6$$
10. 1 sigma and two pi bonds.
11. Nanometre (nm)
12. The amount of nitrogen present in an organic substance and the nitrogen contained in the inorganic compound (ammonia/ammonium).
13. Hydrogen acts as a bridge between two electronegative element.





QUESTION A

Calculate the empirical formula of a compound X with 76.4% carbon and 9.1% hydrogen by mass.

C = $76.4 / 12$	H = $9.1 / 1$	O = $14.5 / 16$	$\frac{1}{2}$ mk
$6.37 / 0.91$	$9.1 / 0.91$	$0.91 / 0.91$	$\frac{1}{2}$ mk
7	10	1	EF = $C_7H_{10}O$ $\frac{1}{2}$ mk

9 Reaction of ketones with Grignard reagent gives tertiary alcohols (1mk)

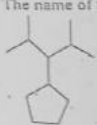
10 The structure below represent

$CH_3-C(=O)-H \rightleftharpoons CH_2=C(OH)-H$ tautomeric isomers (1mk)

11 Calculate the percentage composition of Nitrogen in $CH_3CH_2CH_2CONC(=O)H$

$14 / 15 + 14 + 14 + 28 + 14 + 15 \times 100 = 14 / 100 = 14\%$ $\frac{1}{2}$ mk

12 The name of the structure below is:

 3-cyclopropyl-2,4-dimethylpentane (1mk)

13 Aromatic compounds are different from aliphatic because they cannot be represented by Lewis structures (1mk)

14 List two characteristics of a homologous series-

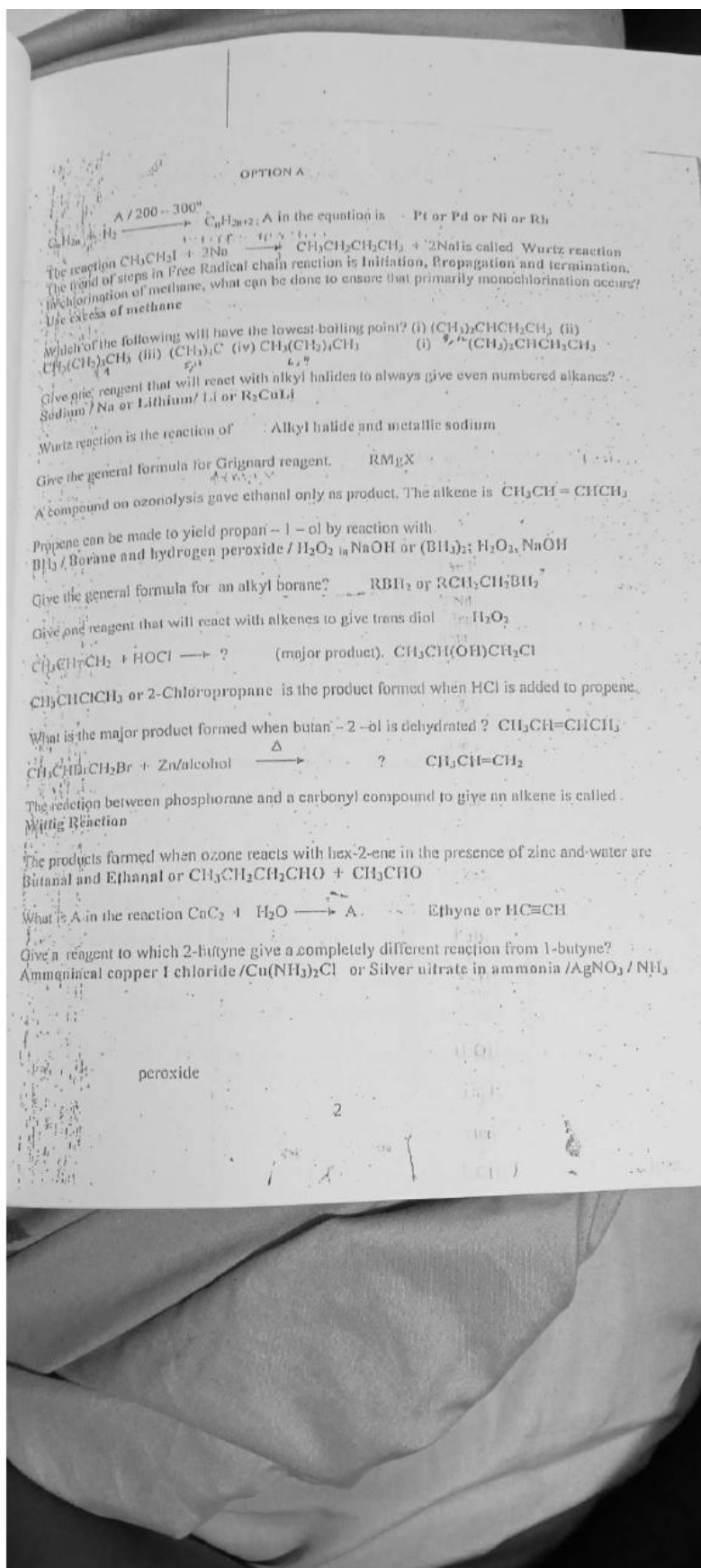
(i) Separated from next by CH_2 (ii) Conform to a general formula (iii) have similar chemical properties (iv) Have gradual change in physical properties (v) Prepared by a general method (any 2 x 1mk = 2mk)

15 How many isomers are possible for the alkane C_6H_{14} 4 (1mk)

16 Give one reagent that can be used to identify terminal alkynes? Ammoniacal copper (I) chloride or Ammoniacal silver nitrate (1mk)

17 Give the characteristics of elements active in odd mass number (1mk)

2



Brohara Tamucipreya Prudent

OPTION A

CHM 121 2013 / 2014 SESSION Answers Mock Eng

OPTION A

Give one property that shows that CH_3OCH_3 and $\text{CH}_3\text{CH}_2\text{OH}$ do not belong to the same homologous series? Have different functional group

Bond breakage that gives free radical is called: Homolytic fission

Inductive effect is due to difference in: Difference in electronegativity

Two reasons for the uniqueness of carbon are:-----

- Carbon has more compounds than all other elements combined;
- Carbon forms long chains (catenation) and ring structures.
- Carbon forms single, double and triple bonds with other carbon atoms or other atoms.
- Carbon has half filled shell in the second period, confers some degree of stability

 $\text{C}_6\text{H}_5\text{Cl}$

This compound can be classified as: Aromatic - non - hydrocarbon

What is an electrophile? Electron loving or electron deficient molecule or reagent

What is the IUPAC name of $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_2\text{CH}_2-\text{CH}(\text{OH})-\text{CH}_2\text{CH}_3$ is: 2,2-dimethyl-6-octen-3-ol

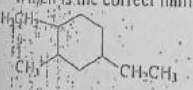
What is the common name of the compound $\text{C}_4\text{H}_9\text{N}(\text{C}_3\text{H}_7)_2$: Dipropyl butylamine

The name of the following compound, $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}_2\text{CHO}$ is: 4-heptenal or Hept-4-enal



is: 2,5-dinitrophenol or o, m'-dinitrophenol

What is the correct name for the compound



4-ethyl-2-methylpropylcyclohexane

What is pyrolysis? Breaking up or cracking a molecule by heating

OPTION A

$\text{C}_6\text{H}_6 + \text{H}_2 \xrightarrow{\Delta, 200-300^\circ} \text{C}_6\text{H}_{12}$; A in the equation is Pt or Pd or Ni or Rh

14 The reaction $\text{CH}_3\text{CH}_3 + 2\text{Na} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 + 2\text{NaI}$ is called Wurtz reaction

15 The first of steps in Free Radical chain reaction is Initiation, Propagation and termination

16 In chlorination of methane, what can be done to ensure that primarily monochlorination occurs? Use excess of methane

17 Which of the following will have the lowest boiling point? (i) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$ (ii) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{CH}_3$ (iii) $(\text{CH}_3)_3\text{CCH}_3$ (iv) $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$

18 Give one reagent that will react with alkyl halides to always give even numbered alkanes. Sodium / Na or Lithium / Li or R_2CuLi

19 Wurtz reaction is the reaction of Alkyl halide and metallic sodium RMgX

20 Give the general formula for Grignard reagent. RMgX

21 A compound on ozonolysis gave ethanal only as product. The alkene is $\text{CH}_3\text{CH}=\text{CHCH}_3$

22 Propene can be made to yield propan-1-ol by reaction with BH_3 / Borane and hydrogen peroxide / H_2O_2 in NaOH or $(\text{BH}_3)_2$; H_2O_2 ; NaOH

23 Give the general formula for an alkyl borane? RBH_2 or $\text{RCH}_2\text{CH}_2\text{BH}_2$

24 Give one reagent that will react with alkenes to give trans diol. H_2O_2

25 $\text{CH}_3\text{CH}=\text{CH}_2 + \text{HOCl} \rightarrow ?$ (major product) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{Cl}$

26 $\text{CH}_3\text{CHClCH}_3$ or 2-Chloropropane is the product formed when HCl is added to propene.

27 What is the major product formed when butan-2-ol is dehydrated? $\text{CH}_3\text{CH}=\text{CHCH}_3$

28 $\text{CH}_3\text{CHBrCH}_2\text{Br} + \text{Zn}/\text{alcohol} \xrightarrow{\Delta} ?$ $\text{CH}_3\text{CH}=\text{CH}_2$

29 The reaction between phosphorane and a carbonyl compound to give an alkene is called Wittig Reaction

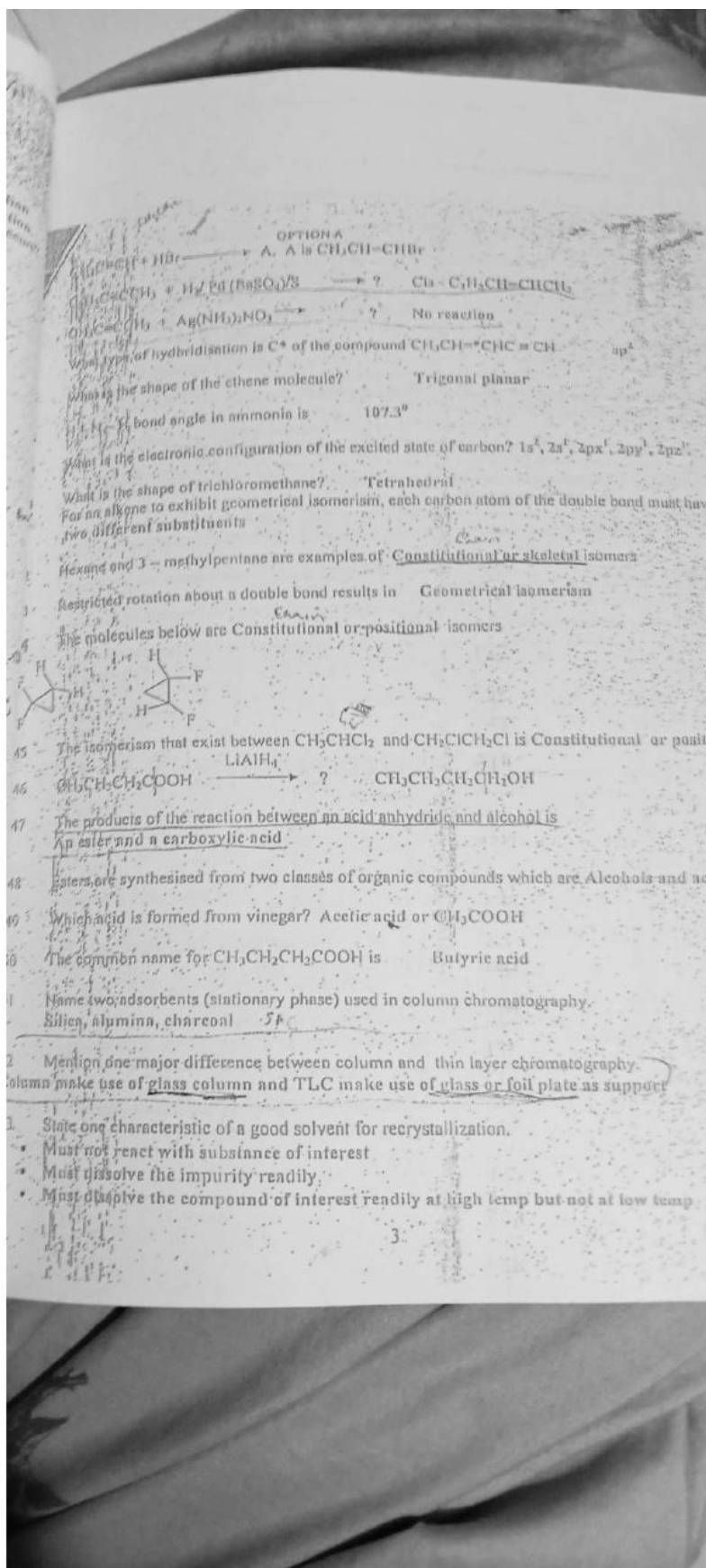
30 The products formed when ozone reacts with hex-2-ene in the presence of zinc and water are Butanal and Ethanal or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO} + \text{CH}_3\text{CHO}$

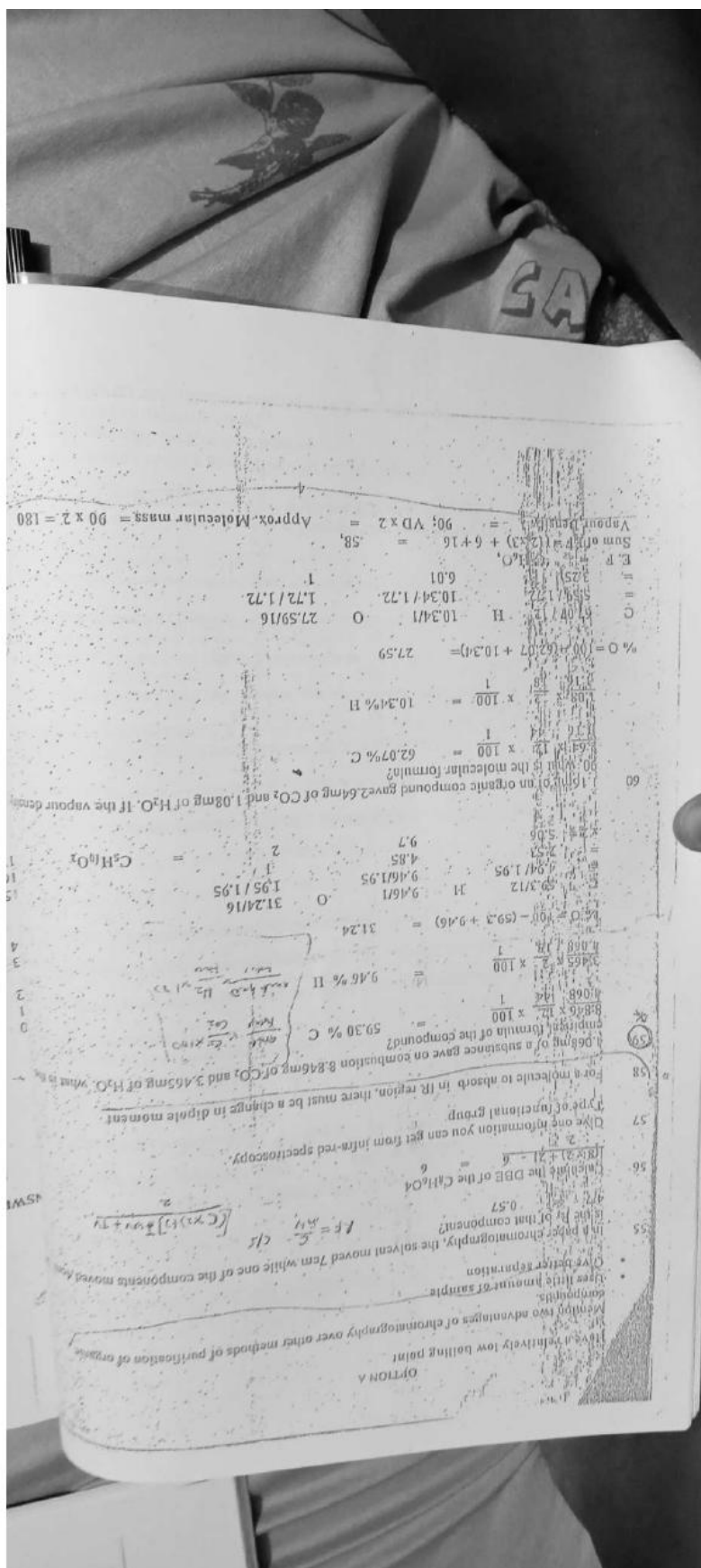
31 What is A in the reaction $\text{CaC}_2 + \text{H}_2\text{O} \rightarrow \text{A}$? Ethyne or $\text{HC}\equiv\text{CH}$

32 Give a reagent to which 2-butyne give a completely different reaction from 1-butyne? Ammoniacal copper I chloride / $\text{Cu}(\text{NH}_3)_2\text{Cl}$ or Silver nitrate in ammonia / AgNO_3

33 peroxide

2

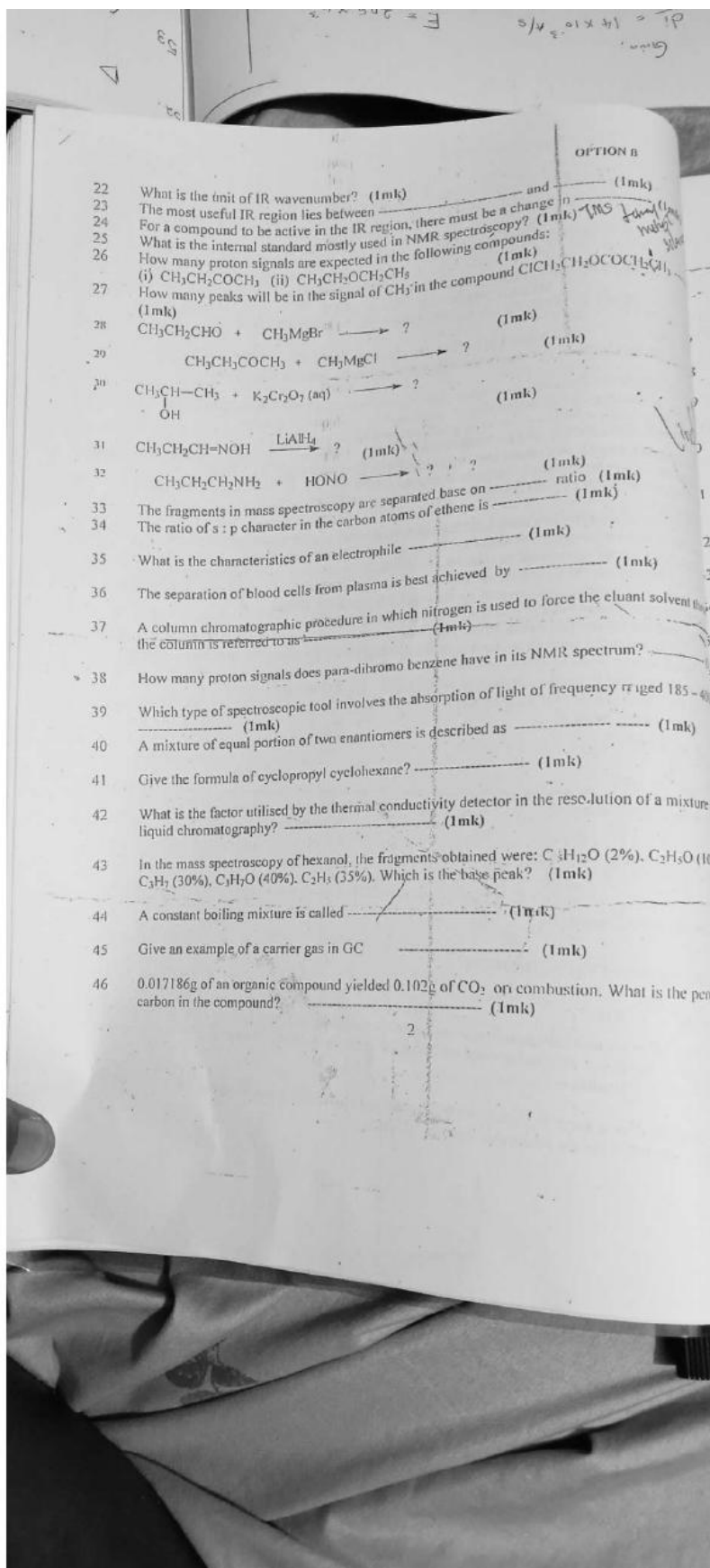


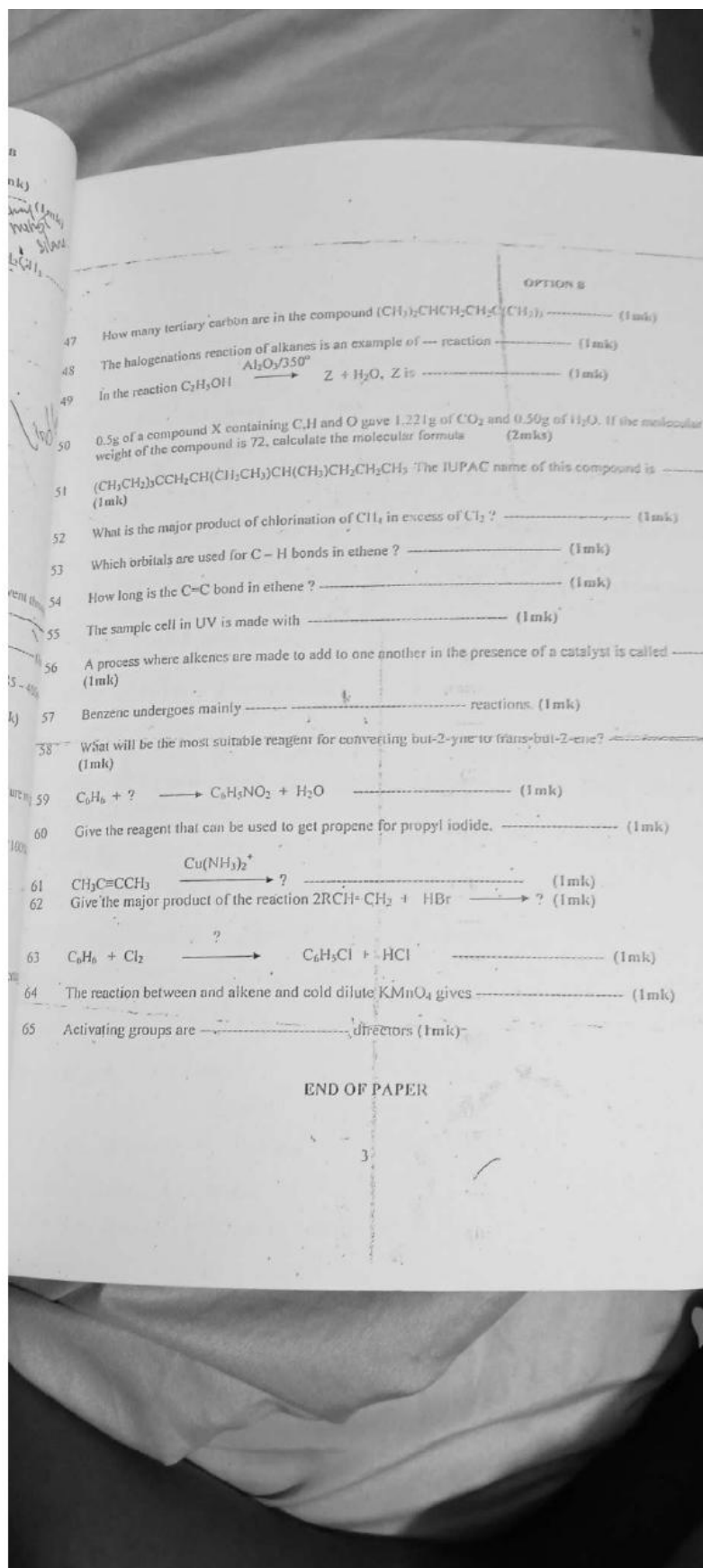


OPTION B
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,
DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATIONS 2014 / 2015 SESSION

CHM 121 ORGANIC CHEMISTRY TIME: 1 ½ HRS OPTION B
ANSWER ALL QUESTIONS Clearly indicate your option on your answer booklet

Friedrich Wohler synthesized _____ by heating ammonium cyanate (1mk)
Two unique characteristic of carbon are: _____ (2mks)
Phenols have _____ group on the benzene ring (1mk)
_____ is used for filtering hot solutions (1mk)
A negative azeotrope is one whose boiling point _____ (1mk)
Separation in chromatography is based on degree of _____ (1mk)
The source of electron in ECD is _____ (1mk)
TCD means _____ (1mk)
The presence of impurity in a compound lead to _____ in melting point. (1mk)
Heterolytic bond cleavage gives rise to _____ (1mk)
A separatory funnel can be used to separate _____ mixture (1mk)
In a TLC procedure, the R_f of a compound was 0.8mm. If the distance moved by the solvent front was 9mm, what distance was covered by the organic compound? (1mk)
Name any four components of a simple distillation set up (2mks)
An organic compound had a distribution ratio between ether and water of 8.50 at 20°C, what weight of it will be extracted from an aqueous containing 0.26g/100ml by using 100ml of ether? (2mks)
Calculate the double bond equivalent of (i) $C_6H_8O_2$ (ii) $C_6H_8O_2Cl_4$ (2mks)
Give the IUPAC name of $(CH_3)_2C=CHCOCH_3$ (1mk)
Give the IUPAC name of _____ (1mk)
18 Give the IUPAC name of _____ (1mk)
19 Give the IUPAC name of _____ (1mk)
20 3.13mg of an organic compound on combustion gave 4.59mg CO_2 and 1.89mg H_2O . 179ml of its vapour at 300K and 750mmHg weighed 0.45g. What is its molecular formula? (2mks)
21 A compound was found to contain 64.0% carbon and 4.03% hydrogen. Its approximate molecular mass was 150. Find the molecular formula. (2mks)





ANSWERS TO ORGANIC CHEMISTRY (CHM121) 2014/2015.

Urea

Uniqueness of Carbon ^{any choice of}
 Ability to catenate i.e. form repeated units and various
 sizes of ring structures.
 It forms single, double and triple bonds with itself and
 other atoms.
 It has half filled shell in the second period which confers
 some degree of stability.
 It readily shares electrons to form millions of covalent compounds.

R-O-H (OH group).

fluted filter paper.

The boiling point of the mixture is maximum. The constant
 boiling mixture has a temperature higher than that
 of the component.

Polarity.

Radioactive beta particle.

TCD: Thermal Conductivity Detector.

decrease in melting point.

Ions

immiscible mixtures.

SOLUTION

R_f of compound = 0.8 mm

Distance moved by solvent = 9 mm.

Distance moved by Organic Compound = ?

$$\text{Retention factor (Rf)} = \frac{\text{Distance moved by Component}}{\text{Distance moved by solvent}}$$

$$0.8 = \frac{D_c}{9}$$

$$D_c = 9 \times 0.8$$

$$= 7.2 \text{ mm}$$

13. 4 components of a simple distillation set up.

- Retort stand
- Condenser
- Bunsen burner
- flask conical
round

$$[4] \text{ Distribution Ratio} = \frac{\text{Molar Concentration in stationary phase}}{\text{Molar Concentration in mobile phase}}$$

$$8.50 = \frac{0.26 \text{ g in } 10 \text{ mL}}{2 \text{ g in } 10 \text{ mL}}$$

$$x = \frac{0.26}{8.50} = 0.030588$$

$$x \approx 0.0306 \text{ g}$$

$$[5] \text{ DBE} = \frac{[(\text{No of Carbon atoms} \times 2) + 2] - (\text{No of Univalent})}{2}$$

$$\text{DBE for } C_6H_5NO_2$$

$$= \frac{14 - 5 + 1}{2} = \frac{10}{2} = 5$$

$$ii) C_6H_5OC_6H_5$$

$$\text{DBE} = \frac{26 - 10}{2} = \frac{16}{2} = 8$$

$(CH_3)_2C=CHCOCH_3 \Rightarrow$ 4-methylpent-3-en-2-one
 or Methylisobutyryl Ketone.

2- aldo benzaldehyde.

4,8-dibromo-6,9-diethyl-3,3-dimethyl-5-dodecenic acid.
 or 4,8-dibromo-6,9-diethyl-3,3-dimethyldodec-5-enoic acid.

4-Chloro-7-ethyl-8-methyl-2-nitrononane.

SOLUTION

3.13 mg of an Organic Compound gave 4.59 mg of CO_2 and 1.89 mg of H_2O .

% Composition of C = $\frac{\frac{12}{44} \times 4.59}{3.13} \times 100$
 (in CO_2 as part of the 3.13mg compound)
 $= \frac{12}{44} \times 4.59 \times \frac{100}{3.13} = 39.994\%$
 $\approx 40\%$

% of H = $\frac{\frac{2}{18} \times 1.89}{3.13} \times 100$
 $= \frac{2}{18} \times 1.89 \times \frac{100}{3.13} = 6.71\%$

% of O = $100\% - (40 + 6.71)\%$
 $= 100\% - 46.71\%$
 $= 53.29\%$

To calculate the empirical formula:

	C	H	O
	$\frac{4.0}{12}$	$\frac{6.71}{1}$	$\frac{58.29}{16}$
	$= \frac{8.33}{8.33}$	$\frac{6.71}{3.33}$	$\frac{3.33}{3.33}$
	1	2	1

$E.F = C H_2 O = 12 + 2 + 16 = 30$
 179ml at 300K and 750mmHg weigh 0.45g.

To calculate volume at S.T.P (760mmHg & 273K)

$$V_2 = \frac{P_1 V_1 T_2}{P_2 T_1} = \frac{750 \times 179 \times 273}{760 \times 300}$$

$$V_2 = 160.75 \text{ ml}$$

$160.75 \text{ ml} = 0.45 \text{ g}$
 $1 \text{ ml} = x \text{ g}$
 $x = \frac{1 \text{ ml} \times 0.45 \text{ g}}{160.75 \text{ ml}} = 0.002799$

molar volume with weigh: 0.002799×22400

Molecular mass = 62.71 g/mol

$(CH_2O)_n = 62.71$
 $30n = 62.71$
 $n = \frac{62.71}{30} = 2.0$

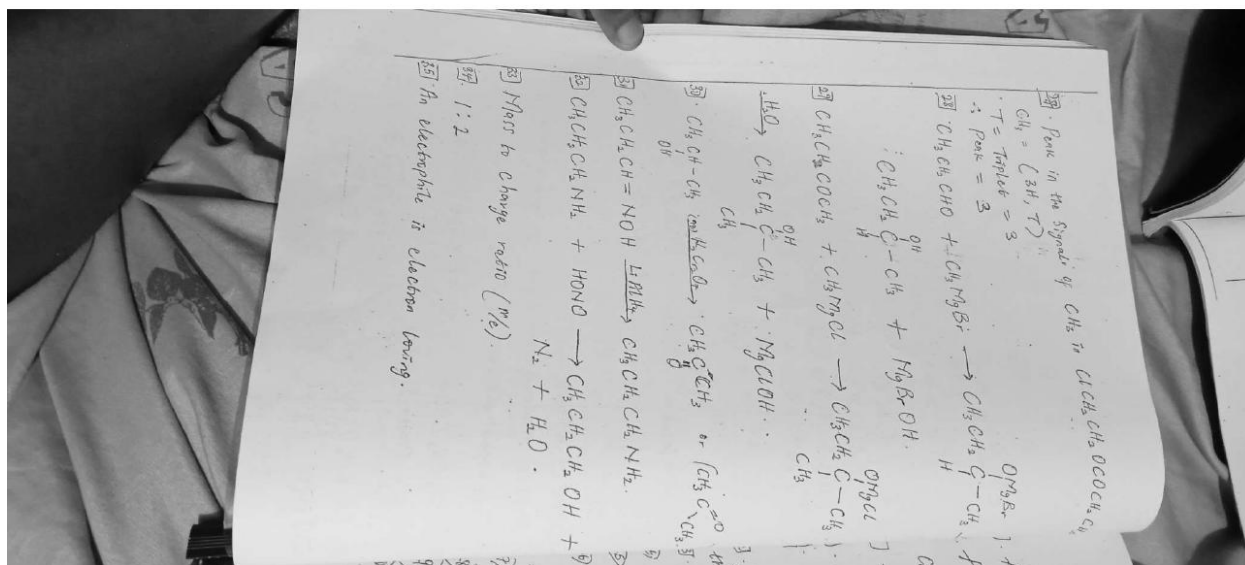
Molecular Formula = $(CH_2O)_2 = C_2 H_4 O_2$

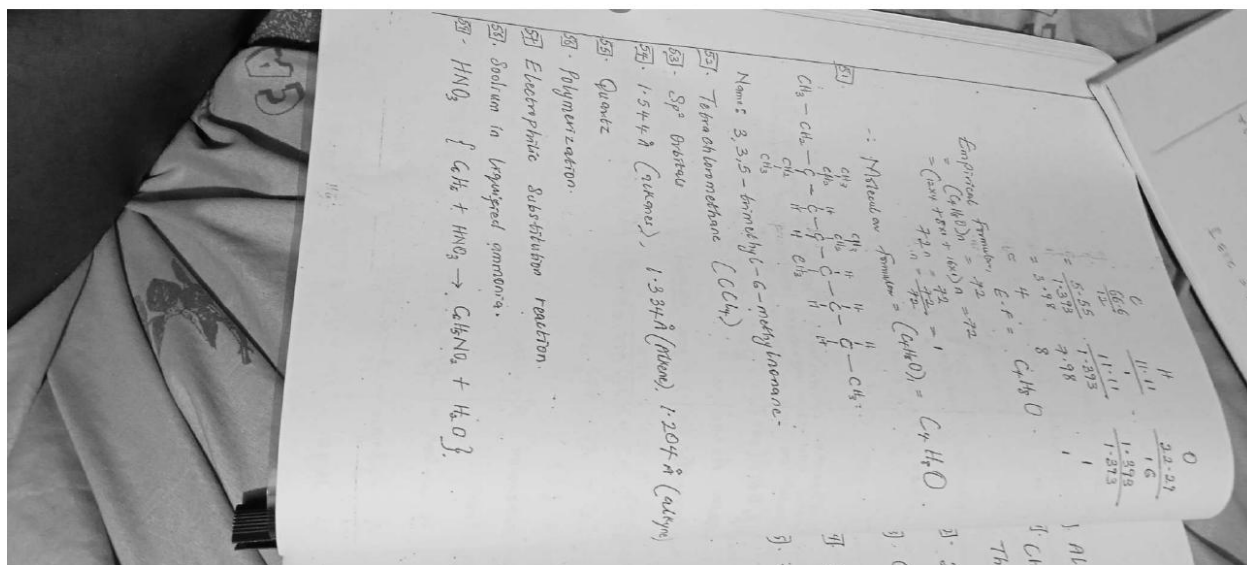
$\% C = 64.0\%$, $\% H = 4.03\%$
 $\% O = 100\% - (64.0 + 4.03)\% = 100\% - 68.03\%$
 $= 31.97\%$

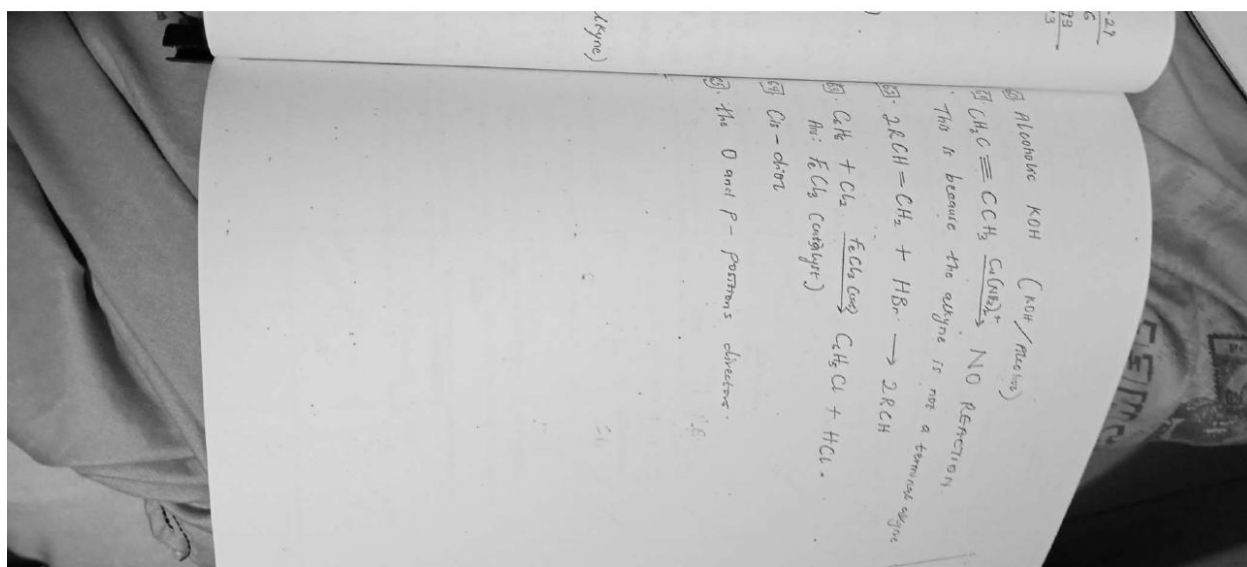
C	H	O
$\frac{64.0}{12}$	$\frac{4.03}{1}$	$\frac{31.97}{16}$
$= 5.33$	4.03	1.998
$\frac{5.33}{1.998}$	$\frac{4.03}{1.998}$	$\frac{1.998}{1.998}$
$= 2.667$	2.067	1
≈ 3	2	1

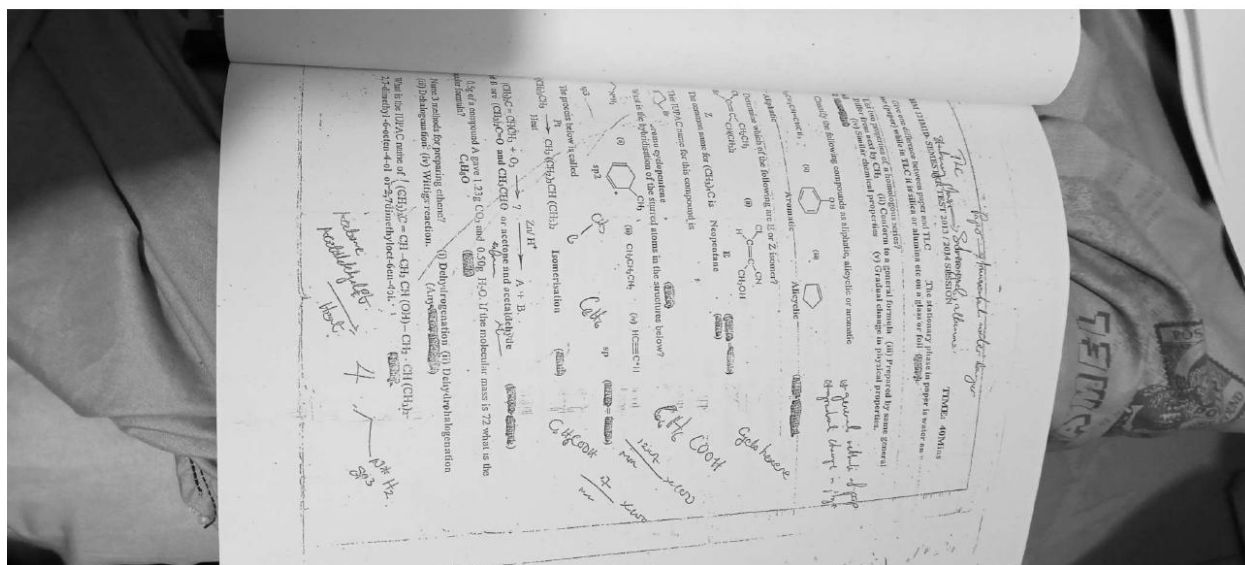
$E.F = C_3H_2O$
 $(C_3H_2O)_n = 150$
 $(12 \times 3 + 2 \times 1 + 16 \times 1)n = 150$
 $(36 + 2 + 16)n = 150$
 $54n = 150$
 $n = \frac{150}{54} = 2.778$
 $n \approx 3.00$
 $\therefore \text{Molecular Formula} = (C_3H_2O)_3$
 $= C_6H_6O_3$

22. cm^{-1}
 23. $400 - 670 \text{ cm}^{-1}$
 24. Dipole moment when light rays are absorbed.
 25. Tetramethylsilane. (TMS)
 26. ① $CH_3CH_2COCH_3 = (3H, T)(2H, Q)(3H, Q)$
 $NMR = \text{Proton Signal} = 3$
 ② $CH_3CH_2OCH_2CH_3 = (3H, T)(2H, Q)(2H, Q)(3H, T)$
 $NMR = \text{Proton Signal} = 4$









Alkyl halides	$R-\overset{\overset{O}{\parallel}}{C}-X$	$-\overset{\overset{O}{\parallel}}{C}-X$ carboxylic acid group	-oyl/halide
Amide (Amides)	$R-\overset{\overset{O}{\parallel}}{C}-NH_2$	$-\overset{\overset{O}{\parallel}}{C}-NH_2$	-amide

